The Biology of Neurotrophins, Signalling Pathways, and Functional Peptide Mimetics of Neurotrophins and their Receptors

Stephen D. Skaper*

Neurology Centre of Excellence for Drug Discovery, GlaxoSmithKline Research & Development Limited, New Frontiers Science Park, Third Avenue, CM19 5AW, Harlow, Essex, UK

Abstract: The neurotrophins are a family of closely related proteins that were first identified as survival factors for sympathetic and sensory neurons, and have since been shown to control a number of aspects of survival, development and function of neurons in both the central and peripheral nervous systems. Limiting quantities of neurotrophins during development control the numbers of surviving neurons to ensure a match between neurons and the requirement for a suitable density of target innervation. Biological effects of each of the four mammalian neurotrophins are mediated through activation of one or more of the three members of the tropomyosin-related kinase (Trk) family of receptor tyrosine kinases (TrkA, TrkB and TrkC). In addition, all neurotrophins activate the p75 neurotrophin receptor ($p75^{NTR}$), a member of the tumour necrosis factor receptor superfamily. Nerve growth factor (NGF), the best characterised member of the neurotrophin family, sends its survival signals through activation of TrkA and can induce death by binding to $p75^{NTR}$. Neurotrophin engagement of Trk receptors leads to activation of Ras, phosphatidylinositol 3-kinase, phospholipase C-γl and signalling pathways controlled through these proteins, including the mitogen-activated protein kinases. Neurotrophin availability is required into adulthood, where they control synaptic function and plasticity, and sustain neuronal cell survival, morphology and differentiation. Preclinical studies point to the therapeutic potential of neurotrophic factors in preventing or slowing the progression of neurodegenerative conditions. Given the difficulties inherent with a protein therapeutic approach to treating central nervous system disorders, increasing attention has turned to the development of alternative strategies and, in particular, small molecule mimetics. This article will provide an overview of neurotrophin biology, their receptors, and signalling pathways, followed by a description of functional mimetics of neurotrophins acting at Trk receptors. Moreover, exciting recent data describing G-protein-coupled receptor transactivation of Trk receptors and their downstream signalling pathways raise the possibility of using small molecules to elicit neuroprotective effects.

Keywords: Neurotrophins, Trk receptor, p75 neurotrophin receptor, glial cell line-derived neurotrophic factor, neuronal signal transduction, kinases, phosphorylation, nerve growth factor mimetics, TrkA mimetics, neuroregeneration.

INTRODUCTION

Cellular diversity in the nervous system evolves from the concerted processes of cell proliferation, differentiation, migration, survival, and synapse formation. In mammals and other vertebrates, soluble peptide growth factors play essential roles in intracellular communication. They exert their effects by signalling through surface membrane receptors that interact with diverse types of intracellular second-messenger systems. Nerve growth factor (NGF) is the founding and best characterised member of the neurotrophin family [1] and was discovered more than half a century ago [2,3] during a search for survival factors that could explain the deleterious effects of target tissue ablation on the subsequent survival of motor and sensory neurons [4]. Hamburger and Levi-Montalcini discovered that cell death occurred normally during neuronal development and that the hypoplasia of neural centers following limb bud removal was associated with increased degeneration of cells in the centers that would normally supply the extirpated limb [2,4]. Significantly, during normal development, neurons die at the time that their processes enter the target. These studies were important because they identified the target as playing a key role in neuronal development. Implanting mouse sarcoma fragments (target) into the body wall of chicken embryos caused enlargement of sensory and sympathetic ganglia and ganglion hyperplasia some distance from the tumour, suggesting that the neoplastic cells released a soluble, diffusible agent that acted on sympathetic and sensory neurons ('nerve growth factor') (Fig. **1**) [3,4]. The key experiment in the entire growth factor field was the demonstration that treatment with anti-NGF antibodies caused the selective elimination of the sympathetic chain [5,6], giving rise to the central model of neurotrophic factor action. There are two unique features of the actions of NGF on neurons as opposed to those of growth factors on other types of cells. First, NGF regulates functions of differentiated neurons, i.e. growth as opposed to proliferation. Second, NGF is synthesised at a considerable distance from the cell body by peripheral tissues or neurons ('targets') that are contacted by axons of the NGF-sensitive neurons. In the periphery, the tissue sources of NGF (and other neurotrophic factors) are typically non-neuronal cells, whereas in the central nervous system (CNS) they are synthesised predominantly by neurons under physiological conditions [7]. During development, a retrograde flow of NGF is established, transporting NGF from the target into the nerve terminal and up the axon to the cell body [8]. Those neurons that establish this flow survive the period of neuronal cell death, while those that do not, degenerate. Once the retro-

*Address correspondence to this author at the Neurology Centre of Excellence for Drug Discovery, GlaxoSmithKline Research & Development Limited, New Frontiers Science Park, Third Avenue, CM19 5AW, Harlow, Essex, UK; Tel: 0044-1279-622350; Fax: 0044-1279-622555; E-mail: Stephen_d_Skaper@gsk.com

1871-5273/08 $55.00+.00 © 2008 Bentham Science Publishers Ltd.

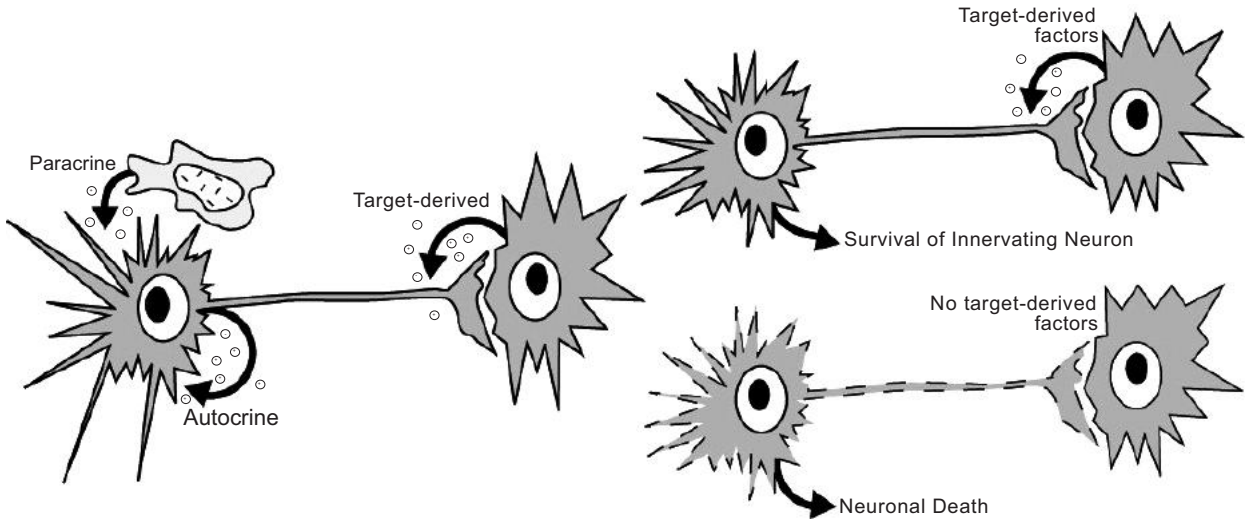


Fig. (1). Target-derived, autocrine and paracrine modes of neurotrophin presentation. During the period of target innervation, neurotrophins support the survival of a restrictive number of neurons expressing the appropriate Trk receptors. Limiting amounts of secreted neurotrophins do not permit the survival of all neurons, which can be rescued by the administration of exogenous neurotrophins.

grade flow of NGF is established, it must continue for the lifetime of the neuron, to maintain the functional differentiated state of the neuron [9]. If the supply of NGF to the target is augmented, some of the neurons that would ordinarily die are rescued [10]. Recent studies on the expression and actions of the NGF family indicate that, in addition to target-derived factor acquisition, autocrine and non-target-derived paracrine modes of factor presentation are likely to be important [11] (Fig. **2**).

Considerable evidence has accumulated over the last fifteen years to indicate that the actions of NGF extend far beyond "classical" effects on cells of the nervous system, to encompass a role for this molecule in the interplay between the nervous, immune, and endocrine systems. Additional, NGF-responsive cells are now known to include lymphocytes [12], mast cells [13], eosinophils [14], other ectodermal-derived cells such as keratinocytes and melanocytes [15,16], and cellular elements of the endocrine system [17]. Expression of the signal-transducing NGF receptor tropomtosin-related kinase (Trk)A on cells of the nervous, immune, and endocrine systems [12,18,19] further strengthens the notion that NGF is a key player in interaction among these systems. These latter aspects of neurotrophic factor biology have been reviewed in detail elsewhere [20,21], and will not be covered here.

One of the major advances of molecular neuroscience over the past several decades has been to recognise that much of the cellular damage resulting from such CNS insults as stroke, trauma, and chronic neurodegenerative disease may be caused by a limited number of endogenously generated molecules with neurotoxic activities. Less well-developed is the idea that endogenous mechanisms exist to provide neuroprotection, and that endogenous molecules may be produced specifically to subserve neuroprotective signalling functions [22]. The importance of the latter class of molecules is illustrated by the severe neurological deficits found in animals in which a neurotrophic protein or its corresponding receptor has been deleted by homologous recombination [23]. Independent lines of investigation have shown the neurotrophins to protect against neuronal dysfunction and death in animal models of injury and neurologic disease [24]. Moreover, clinical studies indicate that certain forms of schizophrenia may be associated with changes in the genetic code of neurotrophic factors [25,26]. These data raise the exciting possibility that neurotrophins may act to protect neurons in disease as well as normal neurons.

BIOLOGY OF THE NEUROTOPHINS

Neurotrophic Proteins

The generality of the phenomenon of programmed cell death after target deprivation (axotomy) has suggested that most neurons respond to and are regulated by neurotrophic factors [27,28]. This hypothesis was validated by the subsequent isolation from pig brain of a second neurotrophic factor, designated brain-derived neurotrophic factor (BDNF), which promoted the survival of several neuronal cell populations not responsive to NGF [29], e.g. placode-derived, but not neural crest-derived sensory ganglionic neurons. Molecular cloning of the BDNF gene [30] revealed its structural similarity to NGF, leading to the concept of the neurotrophin family. Using a homology cloning approach rather than protein purification, two additional members, neurotrophin-3 (NT-3) [31-37] and neurotrophin-4/5 (NT-4/5) [38-40] were later identified (Fig. **3**). The term neurotrophin-4/5 resulted from uncertainties about whether the human neurotrophin-5 [38] was a species homologue of the NT-4 found in *Xenopus* [39]. The nomenclature NT-4/5 was subsequently adopted by a number of investigators to denote the mammalian form of *Xenopus* NT-4, when it was shown that the differences between the two factors were only due to phylogenetic varia-

tion [40,41]. More recently, two novel neurotrophins from the platyfish and carp have been cloned and designated neurotrophin-6 (NT-6) [42] and neurotrophin-7 (NT-7) [43], respectively. These do not have orthologues in mammals or birds and appear to interact with the same receptors as the mammalian proteins. The neurotrophins exhibit actions on distinct, as well as partially overlapping, subsets of peripheral and central neurons [44,45]. Individual neurons may also be responsive to more than one neurotrophin at a given time or at subsequent times during development. Moreover, the actions of neurotrophic factors are not restricted to developing post-mitotic neurons. Indeed, these factors can act on dividing neuroblasts as well as mature neurons that have established stable synaptic contacts. For example, NT-3 enhances the survival of dividing sympathetic neuroblasts as well as young sympathetic neurons [46]. Mature sympathetic neurons subsequently switch their survival dependence to NGF. Thus, the development of certain neuronal populations appears to require the sequential actions of multiple neurotrophins and the coordinated expression of specific neurotrophin receptors [47]. In addition, a number of neuronal populations express neurotrophins. For example, many sensory and sympathetic neurons express BDNF or NT-4/5. Following secretion, NT-4/5 regulates the innervation of sympathetic neurons by preganglionic neuron axons originating from the spinal cord [48,49]. Neurotrophins derived from sensory neurons affect both peripheral and central targets [50]. Within sensory ganglia, BDNF may also have paracrine or autocrine actions.

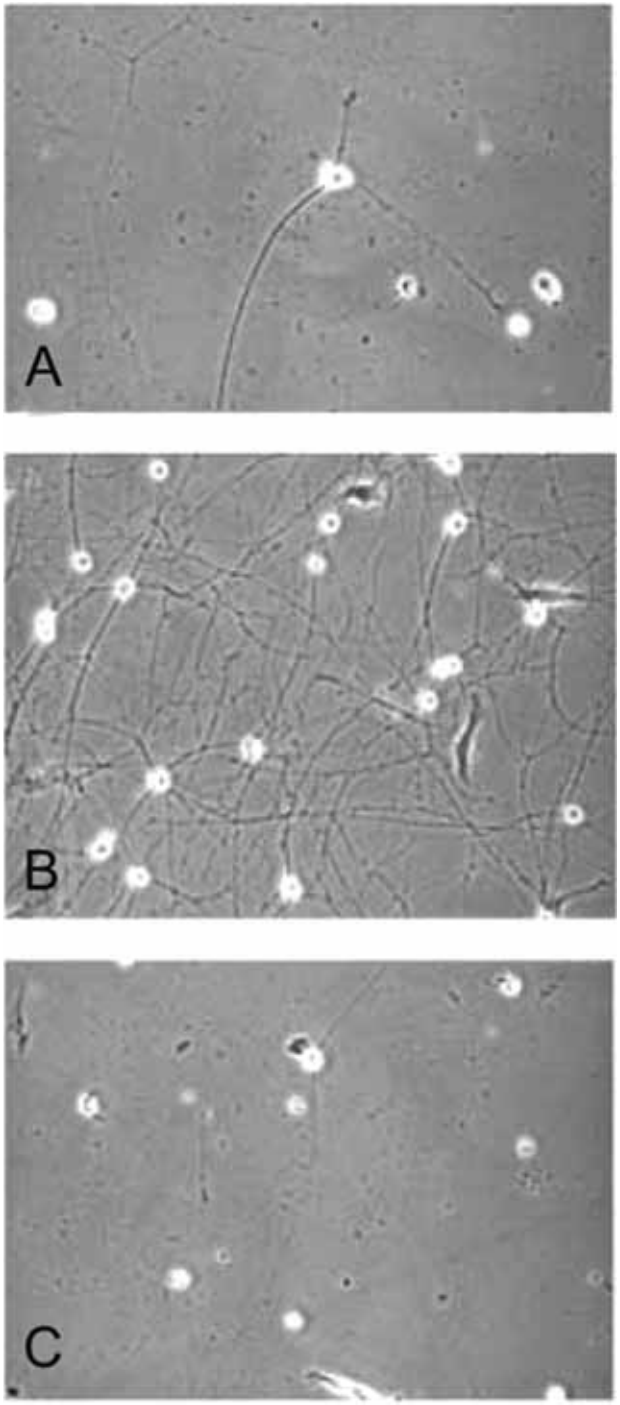


Fig. (2). Chicken embryonic day 10 sensory ganglionic neurons are supported by NGF. Neurons were cultured for 48 hours in the presence of 50 ng/ml recombinant human NGF (**B**), in the absence of NGF (**A**), or with NGF plus anti-NGF IgG (**C**). Note that NGF promotes robust neuronal cell survival and neurite outgrowth, whereas cells cultured without NGF show few neurons and a degenerated neuritic network. Antibodies to NGF effectively neutralised the survival-supporting activity of NGF.

In addition to the neurotrophins, a number of polypeptide factors have been shown to possess neurotrophic activities. These include ciliary neurotrophic factor [51], glial cell line-derived neurotrophic factor (GDNF) [52], insulin-like growth factor [53], and basic fibroblast growth factor (FGF) [54]. Transforming growth factor-β [55] and Sonic hedgehog [56] are other proteins capable of promoting survival of specific CNS neuron populations and protecting these cells from toxic insults.

The neurotrophins and their genes share homologies in sequence and structure, including the existence of multiple promoters [57]. The protein product of each gene includes a signal sequence and a prodomain, followed by the mature neurotrophin sequence. Regulation of neurotrophin maturation may be an important post-transcriptional control point that limits and adds specificity to their actions. Neurotrophin activity is regulated by the proteases responsible for conversion of the proneurotrophins to mature neurotrophins. Interest in these proteases has been stimulated by studies showing that proneurotrophins are secreted from cells and are biologically active [58,59]. Proneurotrophins bind with high affinity to $p75^{NTR}$, and activate signalling pathways controlled by this receptor, which in many cells results in promotion of apoptosis [58,60,61] attributable to concomitant interaction with the sorting receptor sortilin [62], a member of the Vps10p family of receptors. Pro-NGF induces death of oligodendrocytes after spinal cord injury [63] and cortical neurons after injury [64]. Pro-NGF is also abundant in the Alzheimer brain [65], raising the possibility that this form of NGF may contribute to death of septal cholinergic neurons

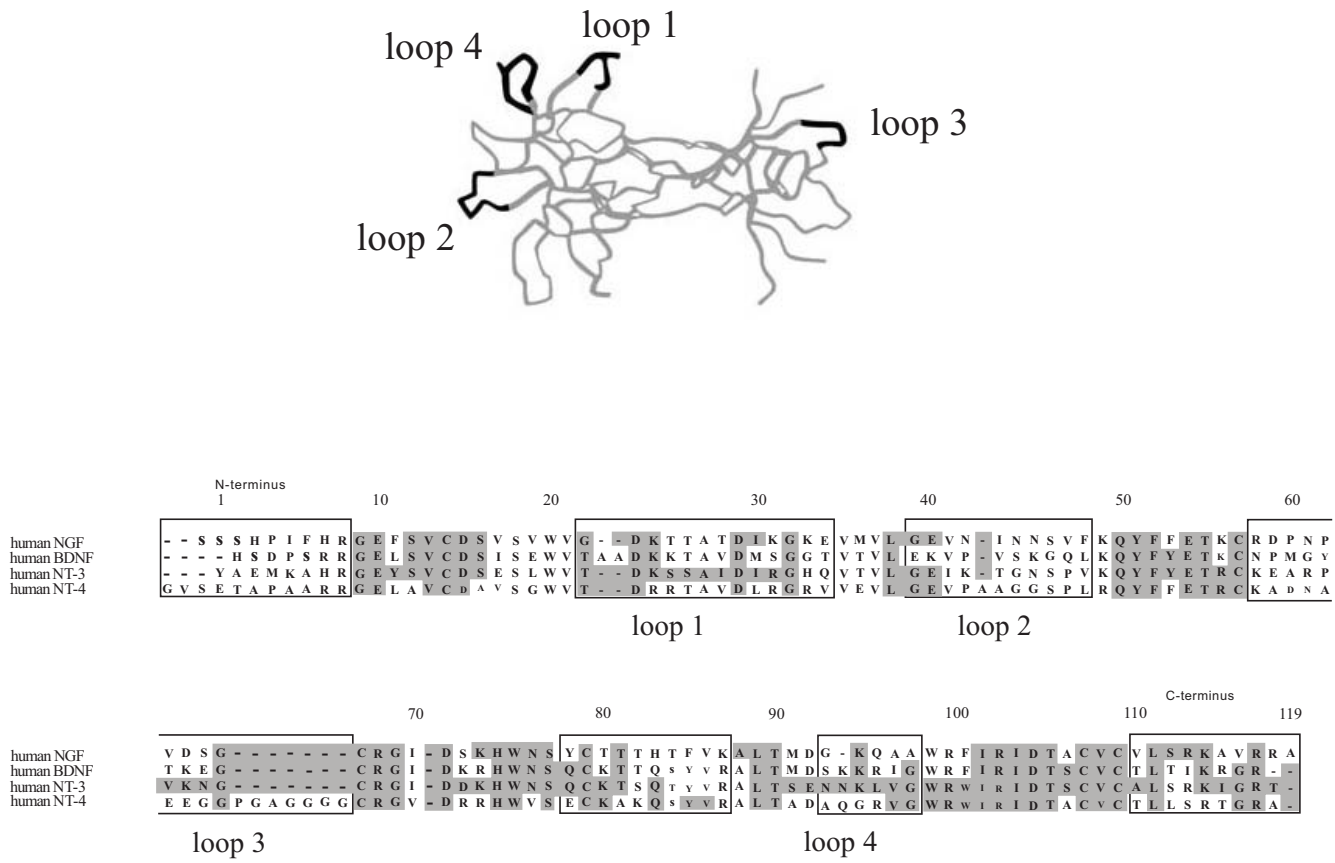


Fig. **(3)**. Structure-based sequence alignment of the neurotrophins. Numbering refers to the sequence of mature human NGF. Positions are numbered from the first residue in each neurotrophin. Note that, because of differences in the lengths of the N-termini of the different neurotrophins, homologous positions in different molecules do not have equivalent numbering. Conserved residues are marked with (*) and regions with low sequence homology [97] are shaded. Dashes indicate gaps introduced for the sake of alignment.

via activation of $p75^{NTR}$.-mediated apoptosis. Indeed, a recent study has shown that pro-NGF produced in the brain under pathological conditions can elicit apoptosis in basal forebrain neurons *in vitro* [66]. In a mouse model for Alzheimer's disease, based on the expression of transgenic antibodies neutralising NGF [67], pathogenesis was hypothesised to result from the antibody (mAb αD11) binding with high affinity to mature NGF, but not pro-NGF, leaving the proneurotrophin free to interact with the $p75^{NTR}$/sortilin complex.

The mature neurotrophin proteins are non-covalently associated homodimers. The structures of NGF, NT-3 and NT-4/5 homodimers and of the BDNF monomer, associated with NT-3 and NT-4/5 monomers, have been solved [68-72]. The neurotrophins share a highly homologous structure and are members of a large superfamily of growth factors that contain a tertiary fold and cysteine 'knot'. These features are present in transforming growth factor-β, platelet-derived growth factor, human chorionic gonadotropin, vascular endothelial growth factor, and others. The cysteine knot consists of three disulfide bonds that form a true knot of the polypeptide chain. Two cysteines that make up the knot are missing from human NT-6. Neurotrophin residues are generally divided into two categories, conserved or variable, based on sequence alignments [45] Amino acid residues implicated in neurotrophin binding that are conserved are likely to represent a common interface to the Trk receptors, while the unique ones may represent elements of specificity [73]. The dimer interface is composed of β-strands that maintain the conformation; these hydrophobic core residues are highly conserved [68,70,72]. In contrast to the β-strands, the β-loops are highly variable (Fig. **3**). Detailed discussions of neurotrophin structure and molecular evolution have appeared elsewhere [52,74-76], and will not be considered here.

Neurotrophin Receptors

The neurotrophins interact with two distinct classes of receptors. The first of these, the p75 neurotrophin receptor ($p75^{NTR}$), is a transmembrane glycoprotein initially identified as a low-affinity receptor for NGF [77,78], but subsequently shown to bind all members of the neurotrophin family with a similar affinity [39,79,80]. $p75^{NTR}$ belongs to the tumor necrosis factor superfamily with an extracellular domain that includes four cysteine-rich motifs, a single transmembrane domain and a cytoplasmic domain that includes a 'death' domain similar to those present in other members of the family [81,82]. While this receptor does not contain a catalytic motif, it interacts with several proteins that relay signals important for regulating neuronal cell survival, differentiation, and synaptic plasticity. Each of the four cysteine-rich repeats of $p75^{NTR}$ participates in binding to NGF [82]. The three-

dimensional structure of the extracellular domain of $p75^{NTR}$ in association with an NGF dimer suggests that binding of NGF to $p75^{NTR}$ may result in dissociation of $p75^{NTR}$ multimers and propose the possibility that Trk and $p75^{NTR}$ monomers simultaneously bind the same neurotrophin monomer. A novel $p75^{NTR}$–related protein, NRH2, lacks the extracellular cysteine-rich repeats present in $p75^{NTR}$ and is unable to bind NGF, but regulates NGF binding to TrkA [83].

In mammals, the Trk subfamily of receptor tyrosine kinases constitute the second major class of neurotrophin receptors. The extracellular domain of each of the Trks consists of a cysteine-rich cluster followed by three leucine-rich repeats, another cysteine-rich cluster and two Ig-like domains (Fig. **4**). Each receptor has a single transmembrane region that terminates in a cytoplasmic, tyrosine kinase-containing domain surrounded by several tyrosine residues that serve as phosphorylation-dependent docking sites for cytoplasmic adaptors and enzymes. The neurotrophins dimerise their cognate Trk receptor, resulting in activation *via* transphosphorylation of the cytoplasmic domain kinases. Specificity of neurotrophin action is believed to be achieved in part by the selective interaction between members of the Trk family of receptors and the different neurotrophins. Thus, NGF binds to TrkA [84-86], TrkB binds BDNF and NT-4/5 with high affinity [87-89], and TrkC binds NT-3 [90]. NT-3 can also interact, albeit with less efficiency, with TrkA and TrkB [41,88,89,91]. A major site at which neurotrophins interact with Trk receptors is in the membrane-proximal Ig-like domain. The three-dimensional structure of this domain in each of the Trk proteins has been solved [92], as well as the structure of NGF bound to the TrkA Ig domain [72]. Differential splicing can also influence neurotrophin interaction with the cognate Trk receptor. In the case of TrkA and TrkB, insertion of short amino acid sequences into the juxtamembrane region of the extracellular domain enhances receptor binding to non-preferred ligands [93-95]. TrkA and TrkB isoforms that lack these inserts are activated strongly only by NGF and BDNF, respectively. Splicing can also generate isoforms of TrkB and TrkC with short cytoplasmic motifs lacking a tyrosine kinase domain. Expression of the non-kinase-containing isoform can inhibit dimerisation of Trk receptors, which in turn inhibits signalling responses of neurotrophins [96]. Truncated Trk receptors are capable of signalling, as evidenced by truncated TrkB-T1 isoform mediating BDNF-evoked calcium release from intracellular stores in glial cells through a G protein and inositol trisphosphate-dependent pathway [97]. Differential splicing of the *trkC* gene results in expression of a TrkC isoform with an amino acid insert within the tyrosine kinase domain, with resulting modification of kinase substrate specificity and interference in ability to promote neuronal cell differentiation [98]. Independently from differential splicing, other factors can modulate neurotrophin binding and action. For example, some of the signalling consequences of TrkB activation following ligand engagement are linked to receptor recruitment to cholesterol-rich lipid rafts [99].

Trk receptor function is modulated by $p75^{NTR}$ on several levels—by promoting ligand binding, by promoting accessibility to neurotrophins through promotion of axonal growth and target innervation, and by promoting endocytosis and retrograde transport to membrane compartments where internal engagement of neurotrophins with Trk receptors may promote signalling. For example, $p75^{NTR}$ inhibits activation of Trk receptors by non-preferred neurotrophins both *in vivo* and *in vitro* [100,101]. The presence of $p75^{NTR}$ potentiates activation of TrkA by suboptimal concentrations of NGF, although it does not appear to potentiate the activation of other Trk receptors similarly by their ligands [102,103], and collaborates with TrkA to form high-affinity binding sites for NGF [104]. In addition to the promotion of binding of NGF to TrkA, $p75^{NTR}$ can promote retrograde transport of several neurotrophins [105]. $p75^{NTR}$ may may reduce ligand-induced Trk receptor ubiquitination, thereby delaying Trk internalisation and degradation [106], or promote Trk receptor endocytosis through polyubiquitnation and subsequent internalisation to endosomal compartments, resulting in enhanced signalling [107]. These findings each suggest a mechanism by which $p75^{NTR}$ may promote axon growth and target innervation *in vivo* and *in vitro* [108,109]. Significant, albeit incomplete, sensory and sympathetic deficits have been observed in mice lacking $p75^{NTR}$ [110-112], although the precise $p75^{NTR}$ function(s) responsible for preventing these deficits remains to be clarified.

Gene knockout experiments for the neurotrophins and the Trk receptors have proven extremely useful in understanding the roles of these ligand—receptor pairs in nervous system development and function [23]. Severe sensory and sympathetic neuropathies occur in mice carrying a disrupted TrkA receptor gene [113], similar to the immunosympathectomy produced by NGF antibodies [4]. Motor neurons, whose numbers are drastically reduced in mice lacking TrkB [114] are not affected in mice lacking both BDNF and NT-4/5 [115,116]. These results suggest that other ligands, perhaps NT-3 act on TrkB *in vivo*. Immunotoxins targeting receptor proteins have also been used to selectively eliminate neuronal populations. For example, a toxin composed of a $p75^{NTR}$ antibody linked to the ribosome-inactivating protein saporin has been used to destroy basal forebrain cholinergic neurons [117,118]. Using another approach, one can construct chimeric proteins formed of a neurotrophin and the catalytic portion of diphtheria toxin to investigate quantitatively what contributions $p75^{NTR}$ and a particular Trk make to a neurotrophin's interaction with neural cells normally expressing one or both of these receptor proteins. One such study utilised a chimera in which the native receptor binding domain of diphtheria toxin was replaced with a synthetic gene encoding rat NT-4/5 (DAB_{389}-NT4) [119]; fusion protein was then applied to induce cytotoxicity in various receptor-expressing cell populations. Neural cell types expressing only $p75^{NTR}$ or $p75^{NTR}$ and full-length or truncated TrkB were used to evaluate the cytotoxic efficacy of DAB_{389}-NT4. The fusion protein produced a concentration-dependent killing of all cell populations with LC_{50} values that largely reflected the known NT-4/5 binding affinities for these receptor proteins. Swiss 3T3 fibroblasts, which lack both $p75^{NTR}$ and TrkB, exhibited >1,000-fold difference in LC_{50} values for DAB_{389}-NT4 and diphtheria toxin (20 ± 3 nM and 80 ± 17 pM, respectively (Fig. **5A**). $p75^{NTR}$ –expressing neuroblastoma neuro-2a cells and hippocampal astrocytes, the latter having as well truncated (but not full-length) TrkB [120], responded to DAB_{389}-NT4 with low nanomolar LC_{50} values, as might be expected for cells lacking a $p75^{NTR}$ –Trk

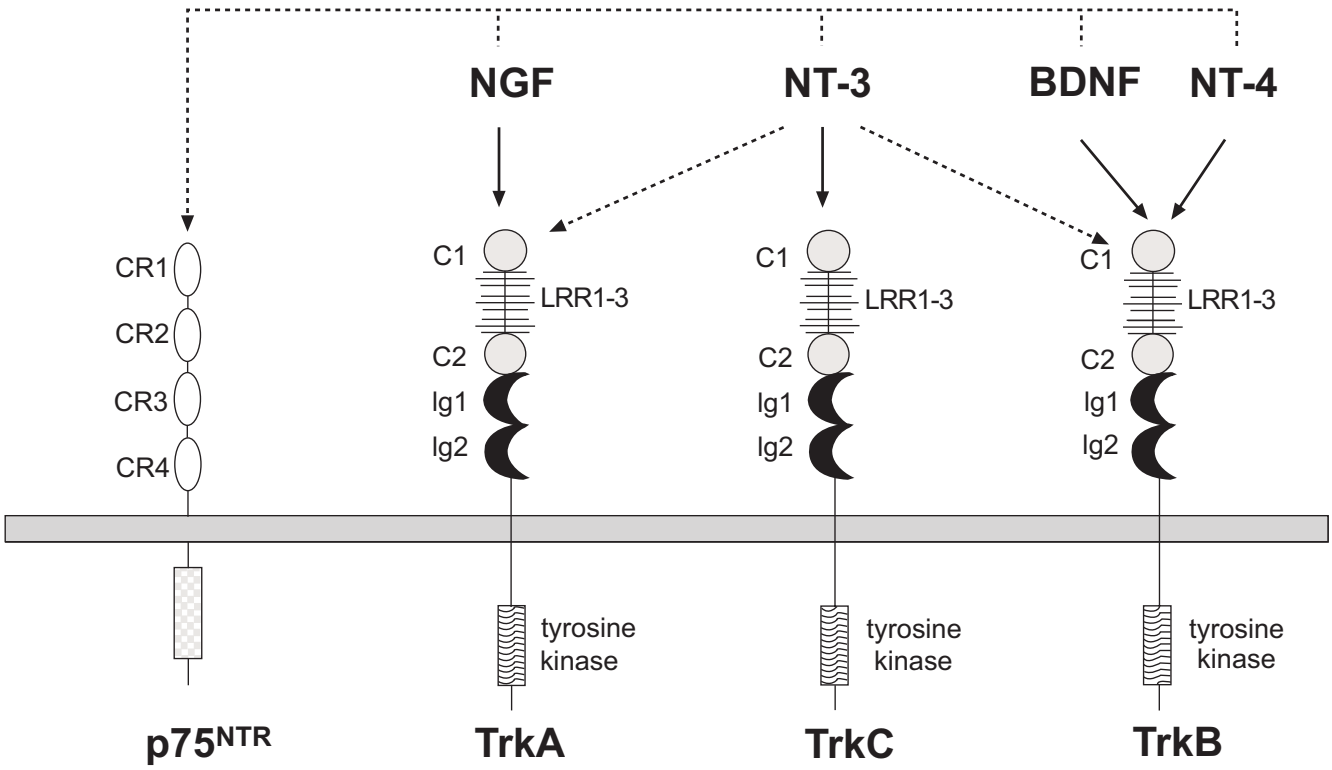


Fig. (4). Neurotrophins and their receptors. The neurotrophins display specific interactions with the three Trk receptors: NGF binds TrkA, BDNF and NT-4 bind TrkB, and NT-3 binds TrkC. In some cellular contexts, NT-3 can also activate TrkA and TrkB albeit with less efficiency. All neurotrophins bind to and activate $p75^{NTR}$. CR1-CR4, cysteine-rich motifs; C1/C2, cysteine-rich clusters; LRR1-3, leucine-rich repeats; Ig1/Ig2, immunoglobulin-like domains.

complex (Fig. **5B**,**C**). Sensitivity of the latter cell types to intact diphtheria toxin, however, remained essentially unchanged. In contrast to the latter cell types, DAB_{389}-NT4 was far more efficacious toward cerebellar granule neurons, killing these cells at concentrations (LC_{50}, 70 $\pm$ 13 pM) (Fig. **5D**) consistent with an interaction of the toxin with high-affinity sites. Cerebellar granule neurons are reported to express $p75^{NTR}$ and TrkB [121] and respond to NT-4/5 and BDNF [122]. The cytotoxic effects of DAB_{389}-NT4 but not diphtheria toxin were fully antagonised by excess NT-4/5 or BDNF (Table **1**), further suggesting that the observed actions of the fusion toxin were indeed mediated by a specific ligand-receptor association [118]. TrkB activation did not appear to be a prerequisite for these effects, however, as inhibitors of Trk tyrosine kinase activity (e.g. K252a) did not diminish DAB_{389}-NT4 cytotoxicity. The data illustrate the feasibility of discriminating NT-4/5 receptor affinities by means of an intact cell system, independent of receptor activation and its downstream trophic consequences, e.g. cell survival or proliferation. Conceivably, it should be possible to design other chimeras fused to the NH_2-terminus of NT-4/5. The same cannot be done for NGF, however, given that a free NH_2-terminus is necessary for NGF receptor binding [123-125].

Neurotrophin Signalling Pathways

Many extracellular signals transduce their cellular responses by regulating tyrosine phosphorylation of their target proteins. Ligand-induced oligomerisation of receptor protein tyrosine kinases and autophosphorylation have been established as a general mechanism for the activation of growth factor receptors, as well as many other families of cell surface receptors [126]. The Trk receptors are typical receptor tyrosine kinases whose activation is stimulated by neurotrophin-mediated dimerisation and transphosphorylation of activation loop kinases [127]. Trk receptors are activated specifically by the mature and not the pro-forms of the neurotrophin gene products [58]. Thus, the proteases that control processing of proneurotrophins control Trk receptor responsiveness. The cytoplasmic domains of the Trk receptors contain several additional tyrosines that are also substrates for phosphorylation by each receptor's tyrosine kinase. When phosphorylated, these residues form the cores of binding sites that serve as a scaffolding for the recruitment of a variety of adaptor proteins and enzymes that ultimately propagate the neurotrophin signal [128]. Within the activated Trk molecule the phosphotyrosines and their surrounding amino acid residues create binding sites for proteins containing phosphotyrosine-binding (PTB) or Src homology 2 (SH2) domains. The major pathways activated by the Trk receptors are Ras, Rac, phosphatidylinositol 3-kinase (PI 3-kinase), phospholipase C-γ1 (PLC-γ1) and their downstream effectors [127,128]. In addition, endocytosis and transfer of Trk receptors to different membrane compartments control the efficiency and duration of Trk-mediated signalling, in part because many adaptor proteins are localised to specific membrane compartments [129].

The Ras-MAP Kinase Pathway

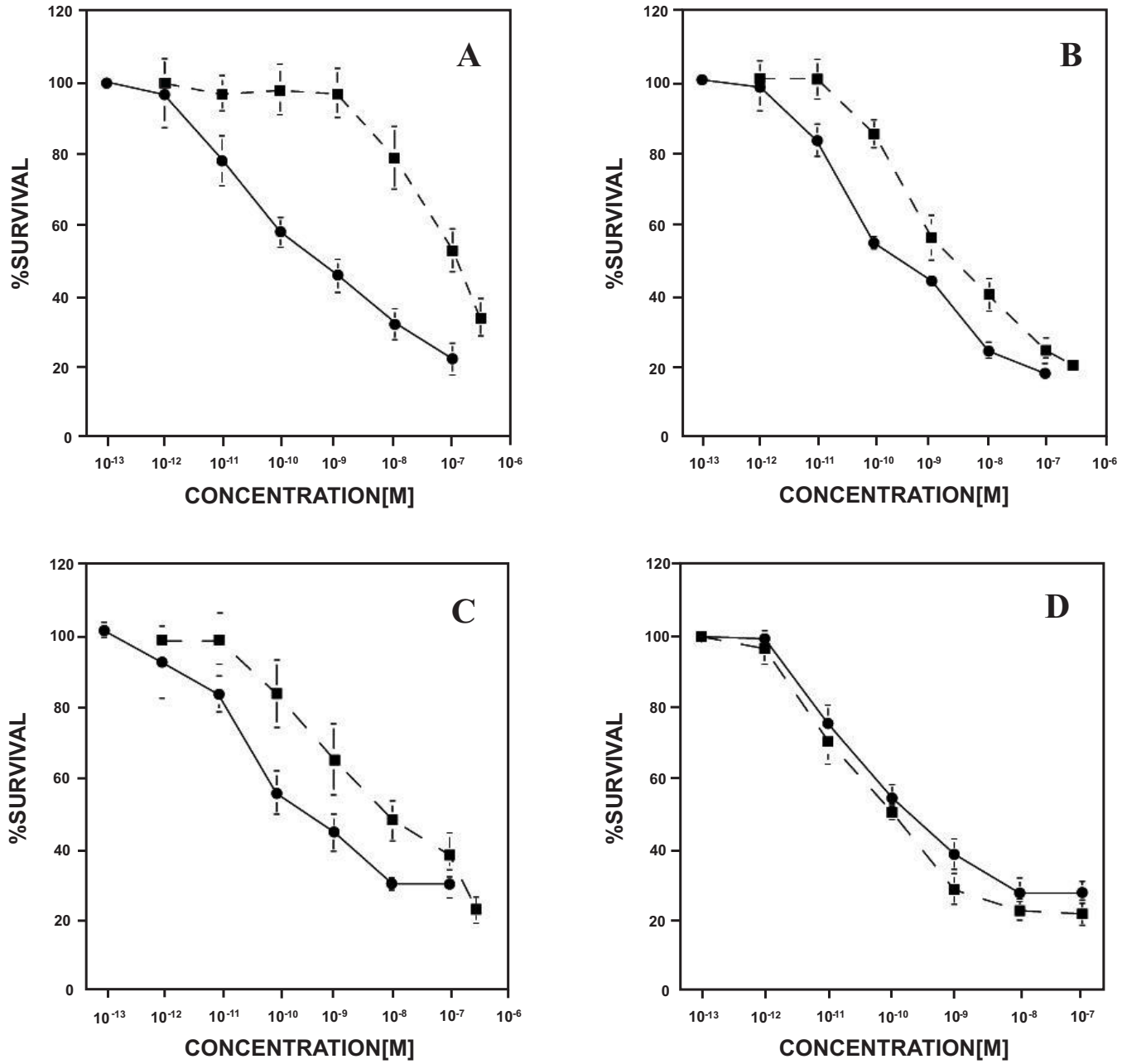


Fig. (5). Cytotoxicity of DAB_{389}-NT4 and diphtheria toxin to (**A**) Swiss 3T3 fibroblasts, (**B**) neuro-2a neuroblastoma cells, (**C**) hippocampal astrocytes, and (**D**) cerebellar granule neurons. Cells were cultured for 72 h (96 h in the case of granule neurons) with the indicated concentrations of DAB_{389}-NT4 (■) or diphtheria toxin (●). Results are given as percentages of survival in control (untreated) sister cultures, as determined by a colourimetric assay for cell number, and are mean + SD. [Reprinted from *Journal of Neurochemistry 68(2),* A Negro and S.D. Skaper, Synthesis and cytotoxic profile of a diphtheria toxin—neurotrophin-4 chimera, 554-563 (Fig. **5**), Copyright (1997), with permission from Blackwell Publishing Ltd].

Activation of Ras is required for normal neuronal cell differentiation and also promotes survival of many neuron subpopulations. Interestingly, $p21^{ras}$ may mediate NGF signals in some neurons (for example, avian sensory or rat sympathetic neurons) but not in others (avian sympathetic neurons), despite the ability of NGF to activate Ras in all Trk-expressing neurons [130-132]. Further evidence for a key role of Ras in NGF-dependent neurons is proposed by the observation that sympathetic and sensory neurons cultured from neurofibromin-1 null mice (neurofibromin inactivates Ras by enhancing its GTPase activity) lose their requirement for NGF [133]. Transient activation of Ras is mediated by the adaptor protein Shc, which is recruited to its recognition site on activated Trk *via* interactions with the Shc PTB domain [134]. This sets in motion the Ras-mitogen-activated protein (MAP) kinase cascade. Trk-mediated phos-

phorylation of Shc creates a phosphotyrosine site on Shc that recruits another SH2 domain-containing protein, Grb2, which is bound to the nucleotide exchange factor SOS (son of sevenless). Activation of Ras may also be facilitated by neurotrophin-dependent phosphorylation and activation of the Ras G protein-releasing factor, RasGRF1 [135]. A Grb2-SOS complex translocates to the plasma membrane, where SOS activates the small G protein p21^{ras} by replacing GDP with GTP [128]. Active Ras stimulates signalling through the protein kinases Raf-1, B-Raf, MAP-ERK kinase kinase (MEK), and p38 MAP kinase [136-139]. Activation of one or more of these kinases leads to the phosphorylation and activation of MEK and the MAP kinase isoforms, ERK-1 and ERK-2 [136,138,140] (Fig. **6**). Introducing activated forms of upstream regulators of MAP kinase, including Ras [141], Raf [142], or MEK [143], mimics NGF by inducing neurite outgrowth in PC12 pheochromocytoma cells. The dominant negative forms of Ras [144], Raf [145], and MEK [143] block the neurite outgrowth in PC12 cells stimulated by NGF. A specific inhibitor of MEK, PD098059, which has no activity at PI 3-kinase or other serine-threonine or tyrosine kinases [146] arrested neurite outgrowth but did not block survival in PC12 cells terminally differentiated with NGF [147]. This observation is consistent with the observation that a MAP kinase-independent pathway mediates the survival effect of NGF on superior cervical ganglion neurons

Table 1. Antagonism of DAB$_{389}$-NT4 Cytotoxicity by Neurotrophic Proteins

Toxin +Competitor	% Survival		
	Neuro-2	Astrocytes	Cerebellar Granule Neurons
DAB$_{389}$-NT4			
Alone	43.8 ± 5.5	43.4 ± 0.4	38.5 ± 4.5
+ NT-4/5	107.0 ± 8.5	99.5 ± 12.0	105.8 ± 5.8
+ BDNF	96.1 ± 2.2	95.0 ± 7.1	108.8 ± 4.5
+ NGF	99.0 ± 5.2	60.4 ± 7.6	ND
+ CNTF	48.5 ± 4.1	39.6 ± 3.4	37.0 ± 7.0
Diphtheria toxin			
Alone	45.0 ± 5.0	34.7 ± 0.6	ND
+ NT-4/5	39.2 ± 0.2	29.8 ± 3.4	ND

Neuroblastoma neuro-2a cells, hippocampal astrocytes, or cerebellar granule neurons were cultured from 3 days with 5 nM DAB389-NT-4 (1 nM for granule neurons) and a 100-fold excess of recombinant rat NT-4/5 or BDNF, mouse NGF, or recombinant human CNTF. Alternatively, cells were incubated with 1 nM diphtheria toxin with or without 100 nM NT-4/5. Survival was expressed relative to values for sham (no toxin)-treated sister cultures (Defined as 100%). Data are mean ± SD (n=8). ND, not determined. [Modified from *Journal of Neurochemistry 68(2),* A Negro and S.D. Skaper, Synthesis and cytotoxic profile of a diphtheria toxin—neurotrophin-4 chimera, 554-563 (Table **2**), Copyright (1997), with permission from Blackwell Publishing Ltd].

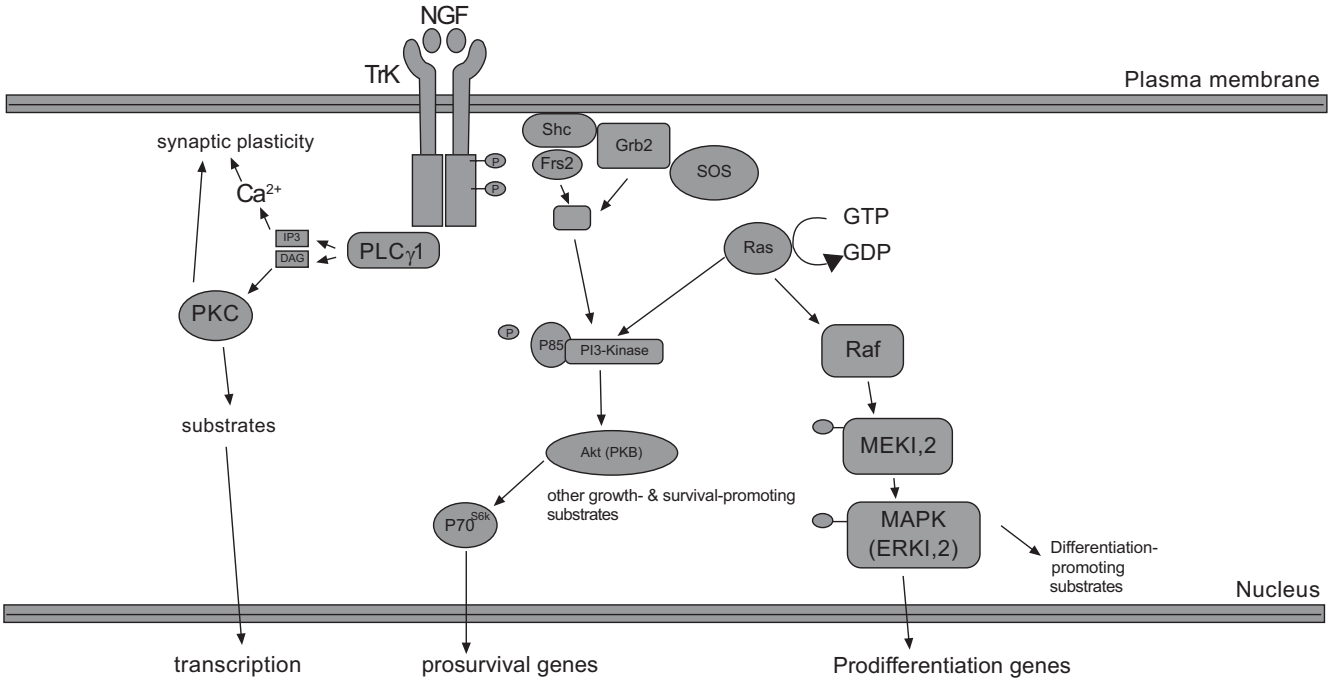


Fig. (6). Neurotrophin signalling. Depicted are interactions of NGF (exemplar neurotrophin) with Trk and major intracellular signalling pathways activated. Neurotrophin binding to Trk receptor leads to dimerisation and autophosphorylation. The linker Shc binds to phospho-Y490 on Trk and to a Grb2-SOS complex. SOS is a nucleotide exchange factor that activates Ras by replacing GDP with GTP. Activated Ras interacts directly with the serine-threonine kinase Raf. The activated Raf leads to the sequential activation of MEK, the mitogen-activated protein kinase-ERK kinase (MAPK). MAPK translocates to the nucleus, where it phosphorylates transcription factors, promoting neuronal cell differentiation. Activation of PI 3-kinase through Ras or Gab1 promotes survival and growth of neurons. Activation of PLC-γ1 results in activation of Ca^{2+}- and protein kinase C-regulated pathways that promote synaptic plasticity. See text for further details.

[148] and serves to illustrate the divergence in neurotrophin signalling. In neurons, p38 is probably activated by a pathway initiated by Ras-mediated binding and activation of the exchange factor Ra1GDS, which results in activation of Ra1 and recruitment of Src [149]. In addition, p38 can be activated through neurotrophin-stimulated activation of the Ras family G proteins, Rin and Rit [150,151]. p38 MAP kinase in turn activates MAP kinase-activated protein kinase-2 (MEKK2). Ras also triggers a signalling cascade through sequential activation of Wnk1, MEKK2 and MEK5 that results in activation of ERK5 (also known as big MAP kinase 1) [152,153]. Two selective p38 inhibitors, SB-202190 and SB-203580, promoted the *in vitro* survival of chicken embryo sensory [147,154], sympathetic, ciliary, and motor neurons [154].

Each of the above MAP kinase cascades has overlapping, as well as distinct targets within the cell [155]. ERK1/ERK2 and ERK5 phosphorylate and activate the ribosomal S6 kinases (Rsks) ($pp90^{rsks}$) [156]. Rsks and MEKK2 each phosphorylate transcription factors such as cAMP-response element-binding protein, which activate transcription of genes essential for differentiation and survival of neurons [156]. In addition to shared targets, the different MAP kinase cascades also have distinct transcription factors as targets [155]. For example, ERK5, but not ERK1/ERK2, activates myocyte enhancer factor 2 [157], while ERK1/ERK2, but not ERK5, activates ELK-1 [158]. The MAP kinase cascades feedback to attenuate and terminate responses through phosphorylation of intermediates and activation of phosphatases. As an example, ERK and Rsk mediate phosphorylation of SOS, which results in dissociation of the SOS-Grb2 complex [159].

The PI 3-Kinase and PLC-γ1 Pathways

Even in rat sympathetic neurons where Ras is mediating survival, it is clear that other signalling pathways are activated, because ERK1 and ERK2 do not mediate the Ras-dependent survival effects [148,160,161]. Generation of P3-phosphorylated phosphoinositides by PI 3-kinase promotes survival of many neurons, and there is evidence to suggest that Ras can signal through PI 3-kinase [162]. In many neurons, Ras-mediated activation of PI 3-kinase initiates the major pathways through which survival signals are conveyed [163]. For example, in PC12 cells a common tyrosine residue(s) may be responsible for coupling Trk to the Ras signalling pathway and to PI 3-kinase [164]. Both couplings appear to occur *via* two common components, Shc and Grb-2, with Grb-2 docking further to pathway-specific proteins (Fig. **6**). Recruitment of Gab1 (Grb2-associated binder-1) by phosphorylated Grb2 permits subsequent binding and activation of PI 3-kinase [165]. In addition, in some neurons, Trk receptor activation results in phosphorylation of insulin receptor substrate-1, which also permits recruitment and activation of PI 3-kinase [166]. PI 3-kinase may mediate neurotrophin survival effects, for example, that of NGF on PC12 cell survival [167]. Pharmacological agents that suppress PI 3-kinase activity block the capacity of insulin-like growth factor-1 [168] and BDNF [169] to sustain the survival of cerebellar granule neurons upon growth signal withdrawal.

The 3-phosphoinositides generated by PI 3-kinase trigger the activation of two protein kinases, the serine-threonine kinase Akt (also known as protein kinase B) and RAC-PK(related to A and C protein kinase) [170], and the p70 ribosomal protein S6 kinase ($p70^{S6K}$) [171]. Akt controls the activities of several proteins important in promoting cell survival, including substrates that directly regulate the caspase cascade, such as BAD, a pro-apoptotic Bcl-2 family member [172]. Phosphorylated BAD is sequestered by 14-3-3 proteins, preventing its pro-apoptotic actions. Akt also regulates the activity of several transcription factors. Through phosphorylation of the forkhead transcription factor, FKHRL1, it promotes association of this transcription factor with 14-3-3 proteins, thereby sequestering it in the cytoplasm and preventing it from activating transcription of several genes whose products promote apoptosis [173]. Experiments with pharmacological inhibitors, as well as expression of wild-type and dominant-inhibitory forms of Akt, demonstrate that Akt but not $p70^{S6K}$ mediates PI 3-kinase-dependent survival [169,174,175].

Targets of PI 3-kinase activation promote axon growth and pathfinding, and neuronal cell differentiation, through P3-phosphorylated phosphoinositides which recruit signalling proteins to the membrane. These include guanine nucleotide exchange factors for Cdc42, Rac and Rho. The latter control organisation of the F-actin cytoskeleton which may account for the ability of neurotrophin gradients to effect growth cone guidance [176].

Trk kinase activation results in PLC-γ1 recruitment to a docking site similar on all Trks. The docked PLC-γ1 is activated through Trk-mediated phosphorylation and then hydrolyses phosphatidylinositol 4,5-bisphosphate to inositol trisphosphate and diacylglycerol. Inositol trisphosphate triggers mobilisation of Ca^{2+} cytoplasmic stores while diacylglycerol stimulates diacylglycerol-regulated isoforms of protein kinase C. These two signalling molecules activate numerous intracellular enzymes, including all protein kinase C isoforms, Ca^{2+}-calmodulin-dependent protein kinases and other Ca^{2+}-calmodulin-regulated targets. PLC-γ1 activation results in activation of, e.g. protein kinase C-δ, which mediates NGF-promoted activation of MEK and ERK1/ERK2 [177]. The activity of PLC-γ1 has been implicated in the ability of TrkB receptors to modulate synaptic transmission and long-term potentiation [178,179].

Other Ras-independent pathways have been implicated in neurotrophin-induced survival and cessation of proliferation. A molecule termed SNT (suc-associated neurotrophic factor-induced tyrosine-phosphorylated target) is inducibly phosphorylated within one minute of NGF addition to PC12 cells [180]. Once phosphorylated, SNT translocates to the nucleus, where it may function as a transcription factor. SNT phosphorylation is not triggered by agents like epidermal growth factor, which fails to induce withdrawal of PC12 cells from the cell cycle. One possibility is that SNT functions as a transcriptional regulator of genes that control cell cycle withdrawal [180]. Distinct signalling pathways mediate activation of SNT and the PI 3-kinase response [181].

Trk Receptor Localisation and Transport: Role in Signalling

Localisation of Trk receptors in specific neuronal subdomains is an important factor in directing cellular commu-

nication in developing and mature neurons. For example, TrkB at synaptic sites may mediate enhancement of glutamate release [182], as well as enhance NMDA receptor function [183]. On the cell surface, Trk receptors become concentrated in caveoli-like domains [184]. Recruitment by activated Trk receptors of signal-transducing proteins on the cell surface appears critical. The endocytosis of Trk receptors after ligand binding, which occurs by both clathrin-dependent and -independent mechanisms, as well as pinocytotic pathways [185], is a crucial factor for the cellular localisation of activated Trk and in recruiting and activating signalling intermediates, such as Shc (Src homology and collagen homology) [186]. The internalised neurotrophin-Trk complex is sorted to an endosomal compartment and then enters into recycling, retrograde transport, or degradation pathways [187]. Internalisation of Trk receptors may be regulated by ubiquitination, suggesting that the differential expression of ubiquitin E2 and E3 ligases may result in differences in Trk receptor trafficking in different populations of neurons.

Lipid rafts are membrane microdomains rich in cholesterol and sphingolipids are important regions for cellular signalling, bringing together transmembrane receptors with adaptor and signalling proteins [188]. Trk receptors are known to be localised in these membrane domains [184,189]. A recent study reported that TrkB specifically localised to intracellular lipid rafts in cortical and hippocampal membranes in response to BDNF, and was dependent on the tyrosine kinase Fyn [190]. BDNF stimulation also resulted in the association between endogenous TrkB and Fyn. In support of a role for lipid rafts in neurotrophin signalling, inhibiting TrkB translocation to lipid rafts, either by using Fyn knock-out neurons or lipid raft-disturbing agents, prevented the full activation of TrkB and downstream PLC-γ [190].

Neurotrophins as Therapeutic Agents

The capability of neurotrophic factors to regulate neuronal cell survival in the developing nervous system and also to promote survival after injury or protect neurons in toxin-mediated disease models in animals has spurred the idea that such proteins could be engaged for the treatment of neurodegenerative disease. For example, the traditional perspective of applying neurotrophins in the context of Alzheimer's disease is based on the premise that neurotrophins are capable of upregulating cholinergic function and of rendering neurons less vulnerable to certain processes causing degeneration [191,192]. For example, NGF maintains and regulates the cholinergic phenotype of basal forebrain neurons [193] *via* retrograde transport of a NGF/TrkA signalling complex from the cortex and hippocampus, where NGF is produced, to the basal forebrain [194]. NGF protects cholinergic neurons from experimental surgical lesions [191,192,195] and ameliorates cholinergic neuron atrophy and spatial memory impairment in aged rats [196-198]. A transgenic mouse model of anti-NGF antibody expression (AD11 mice) is characterised by atrophy and loss of basal forebrain cholinergic neurons, β-amyloid peptide deposition, and cognitive deficits [67].

Another case is that of GDNF, which acts on dopaminergic neurons. Given intracerebrally, GNDF has beneficial effects in toxin-induced rodent and non-human primate models of Parkinson's disease [199-201]. In spite of such encouraging preclinical data, clinical studies have been largely negative. Neurotrophic proteins have failed to show a positive effect in human trials of amytrophic lateral sclerosis [202-206], and in some cases severe side effects resulted [207,208]. A failed outcome was also reported for randomised, controlled trials of GDNF in Parkinson's disease patients [209,210]. Neurotrophic factor treatment of CNS diseases is especially complex, as these polypeptides have poor pharmacokinetics and bioavailability and do not cross the blood-brain barrier to any appreciable extent. Neurotrophic proteins can be modified to increase blood-brain barrier transport [211]: recent studies show that intravenous administration of BDNF, conjugated to an anti-transferrin antibody, is capable of traversing the blood-brain barrier in rats and providing neuroprotection in focal transient brain ischemia [212,213]. Toxicology studies in primates showing cerebellar degeneration following the administration and subsequent withdrawal of high doses of GDNF raise concerns about dosing and targeting of delivery. Viral vector or cell-based gene therapy approaches may offer advantages for the targeted delivery of lower doses of neurotrophic factors. Dopaminergic neuroprotection by viral-mediated delivery of GDNF or the GDNF-related factor neurturin [214] has been demonstrated in animals [199,215,216]. In the case of spinal cord injury, the transplantation of genetically modified cells (e.g. viral expression of neurotrophins either *in vivo* or *ex vivo* in stem cells) to lesion sites has been investigated as a means to augment axonal regeneration [217-219]. A cell-based approach for NGF delivery claimed to shown some benefit in a phase I trial in Alzheimer's disease [220].

Peripheral nervous system (PNS) disorders conceivably are more amenable to a neurotrophic protein therapeutic. NGF, which is selectively trophic for small fibre sensory and sympathetic neurons, represents a potential therapy for diabetic polyneuropathy because of the serious consequences associated with degeneration of those neuronal populations in this condition. Reduced NGF availability may contribute to the pathogenesis of diabetic neuropathy [221,222]. Animal models of neuropathy respond favourably to the exogenous administration of NGF [223]. Two separate phase II clinical trials suggested that recombinant human NGF administration was effective in ameliorating the symptoms associated with both diabetic polyneuropathy [224] and HIV-related neuropathy [225]. These early studies, however, revealed that painful side effects were dose-limiting for NGF. A large-scale phase III clinical trial of 1019 patients randomised to receive either recombinant human NGF or placebo for 48 weeks failed to confirm the earlier indications of efficacy [226]. Such side effects are not unexpected, as increased NGF content in peripheral tissues can lead to inappropriate axonal sprouting [227] and hyperalgesia [228]. Other protein factors with peripheral nerve regeneration activity are efficacious in experimental diabetic neuropathy, and include sonic hedgehog protein [229] and ciliary neurotrophic factor [230]. Whether or not the latter would produce the desired trophic effects of NGF in man without causing hyperalgesia remains to be seen.

Functional Mimetics of Neurotrophins Acting at Trk Receptors

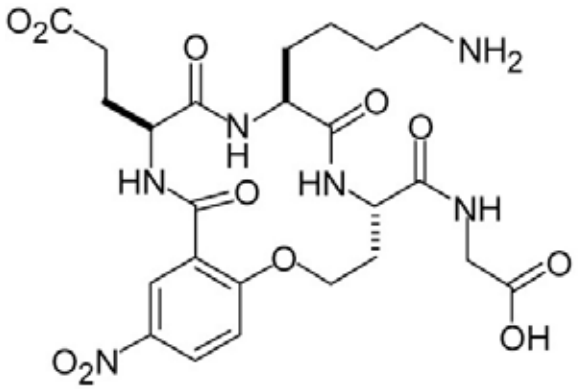
The therapeutic potential of neurotrophic proteins notwithstanding, inherent drawbacks with the use of polypeptides applied as drugs are evident: *in vivo* instability; unsuccessful tissue targeting; pleiotrophic effects due to binding multiple receptors and co-receptors, and activation of multiple pathways that were not intentionally targeted; relatively high production cost for medicinal applications. Such problems may be averted by developing highly selective receptor ligands, with chemical and pharmacological properties such as proteolytic stability, small size, and ability to penetrate tissue barriers. Depending on which region of the polypeptide of interest is mimicked, and where in the receptor the mimetic binds, the small molecule analogues can act as receptor modulators, biological response modifiers, antagonists or partial agonists The pharmacological advantages of mimetics over protein therapeutics are well-documented, and include lack of immunogenicity, low molecular mass, low manufacturing costs and good pharmacokinetic profiles [231].

Numerous functional receptor-ligand interactions occur *via* contact at a few key regions or "hot spots", rather than extensive interactions over the whole protein surface [232,233]. Many of the hot spots at the ligand-receptor interfaces adopt β-turn conformations, especially in the case for proteins of the immunoglobulin superfamily [233-235], and for various growth factors and adhesion molecules [236,237].

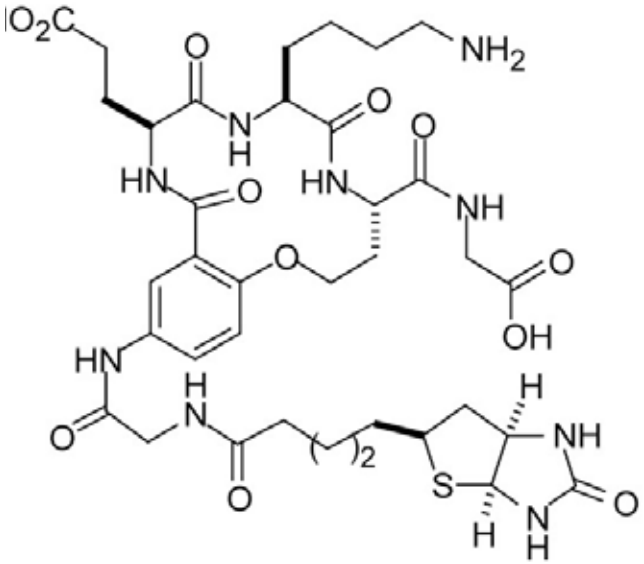
The first rationally designed TrkA agonists were based on the use of ligand mimicry and antibody mimicry strategies [231,238] to generate peptide analogues of two agonists directed to the extracellular domain of TrkA: the peptide sequence of NGF itself [239-241], and the anti-TrkA monoclonal antibody (mAb) 5C3 [242]. This antibody most likely acts through its two variable domains to physically join together two Trk monomers. TrkA binding is mediated by a discrete β-turn region (amino acid residues 92-97) of these ligands, since crystallised peptides and peptidomimetic compounds incorporating these residues were found to activate TrkA and demonstrate neurotrophic effects in cell-based models [243-245]. Such cyclic peptide dimeric ligands may act as TrkA partial agonists, inducing receptor dimerisation, phosphorylation and internalisation [243]. Other conformers or linear peptides were inactive. In fact, most monomer cyclic peptide β-turn mimetics of NGF and BDNF are reported to have only antagonistic properties, inhibiting neurotrophin binding and activity [239,246]. Similar findings show that a dimeric (but not monomeric) cyclised version of amino acids 92-96 of NGF also acts as a partial TrkA agonist, with neurotrophic effects being blocked by TrkA inhibitors [244].

Based on the pharmacophores of the mAb 5C3 and NGF peptide analogues described above, a focused β-turn peptidomimetic library of ~60 compounds was synthesised, leading to the identification of D3 [245], a small, selective, and proteolytically stable agonist of the TrkA receptor (Fig. **7**). A similar compound, C59, lacked neurotrophic effects. These results indicate that a small β-turn peptidomimetic can activate a tyrosine kinase neurotrophin receptor that normally binds a relatively large protein ligand. Although D3 is still far short of an NGF mimetic in terms of its potency, treatment of cognitively impaired aged rats with D3 has been shown to act as a neuroprotective agent. Aged rats showed improved memory and improved cholinergic phenotype. The effect was long-lived, for three months after drug delivery. D3 performed equally or better than NGF [247].

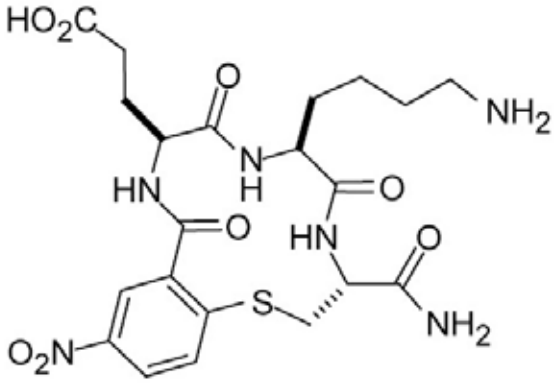
Small, dimeric peptides designed to mimic a pair of solvent-exposed loops important for the binding and activation of the BDNF receptor, TrkB, have been described [248]. The monomer components that make the dimers were based on a monocyclic monomeric peptide mimic of a single loop of BDNF (loop 2) shown previously to be an inhibitor of BDNF-mediated neuronal cell survival [249]. These bicyclic dimeric peptides behaved as partial agonists with respect to



D3



D3-Biotin



C59

Fig. (7). Structures of D3, D3-biotin and C59. [Reproduced from *Molecular Pharmacology, 57(2),* S. Maliartchouk, Y. Feng, L. Ivanisevic, T. Debeir, A.C. Cuello, K. Burgess and H.U. Saragovi, A designed peptidomimetic agonistic ligand of TrkA nerve growth factor receptors, 385-391 (Fig. **1**), Copyright (2000), with permission from the American Society for Pharmacology and Experimental Therapeutics; http://www.molpharm.aspetjournals.org/].

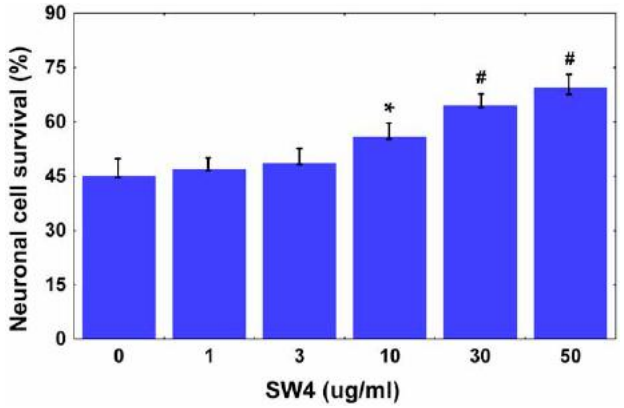
BDNF, promoting sensory neuron survival *in vitro*. A highly conformationally constrained tricyclic dimeric peptide, in which the monomeric subunits were linked by two dimerising linkages retained partial BDNF agonist action, while being 10-fold more potent in its neuronal survival effect than the best of the bicyclic dimers - making it ~2-fold less potent than BDNF itself. Partial agonistic behaviour could result from the compound being less efficient than BDNF in bringing about dimerisation of BDNF or because of inappropriate conformational flexibility of the dimerising linkages (although the retention of partial agonism by the highly constrained tricyclic compound would argue against this). A consequence of reduced dimerisation efficiency could be a reduced ability to cause the autophosphorylation of the five tyrosine residues in TrkB (resulting in reduced activation of downstream signalling pathways). An alternative mechanism for the partial agonist behaviour of the dimeric peptides is that they act through TrkB only, even as full TrkB agonists, and not $p75^{NTR}$. A series of dimeric β-turn mimetics that act as agonists of TrkC have recently been described [250] (Fig. **8**), as well as a tandem repeat cyclic peptide agonist of TrkB [251]. The last approach will be described in more detail elsewhere in this issue.

Cell adhesion molecules such as the neural cell adhesion molecule (NCAM) and N-cadherin are abundantly expressed in the CNS and play an important role in development of the nervous system, e.g. cell migration, axonal growth and guidance, and synapse formation [252]. In the adult cell adhesion molecules modulate neural plasticity involved in learning and memory. NCAM and N-cadherin engage in homophilic interactions both in *cis* and *trans*. Although NCAM and N-cadherin have no known inherent intracellular catalytic activities capable of directly activating any signalling cascades, heterophilic interactions with a variety of binding partners, one of these being the FGF receptor, can function as an activator of intracellular pathways resulting in neurite outgrowth [253-256]. The synthetic NCAM mimetic peptide C3 is a ligand of the NCAM IgI module, which was identified by combinatorial chemistry [257,258]. Addition of the dendrimeric version, C3d (composed of four monomers coupled to a lysine backbone) induced neurite outgrowth from primary neurons through signalling pathways identical to the pathways activated by homophilic NCAM binding [257,259]. In the case of N-cadherin, peptide mimetics of two linear sequences from extracellular domain 1 (HAVDI and INPISGQ) of N-cadherin function as highly specific antagonists [260], while cyclic peptides containing a tandem repeat of the individual motifs behave as N-cadherin agonists and stimulate neurite outgrowth in a manner similar to native N-cadherin [261], presumably by effecting FGF receptor dimerisation. Intriguingly, NCAM and N-cadherin mimetic peptides are also capable of promoting the survival of primary CNS neurons. C3d increased survival of dopaminergic and cerebellar granule neurons induced to undergo apoptosis [261,263], although there appear to be differences in the signalling events taking place following NCAM stimulation leading to neurite extension as compared with NCAM stimulation leading to neuroprotection [263]. With N-cadherin mimetics, the cyclic peptide N-Ac-*CHAVDINGHAVDIC*-NH_2 (SW4) containing the HAVDI sequence in tandem was efficacious in promoting the *in vitro* survival of several populations of CNS neurons in paradigms where FGF-2 is effective [264] (Fig. **9**). The neuroprotective effects of SW4 were inhibited by a monomeric version of the same motif and by a specific FGF receptor antagonist. These data suggest that SW4 functions by binding to and clustering N-cadherin in neurons and thereby activating an N-cadherin/FGF signalling cascade, consistent with the ability of cell adhesion molecules to activate an FGF receptor-signalling cascade in growth cones [253].

Transactivation of Trk Neurotrophin Receptors by G-Protein-Coupled Receptor Ligands

Transactivation of receptor tyrosine kinases by G-protein-coupled receptors (GPCRs) is now recognised as an important signalling mechanism allowing the cell to respond to a vast array of extracellular stimuli [265,266]. This mechanism provides a link between the large family of cell surface GPCRs and tyrosine phosphorylation events to produce a wider spectrum of cellular responses. Transactivation of receptor tyrosine kinases is believed to be critical for physiological functions. For example, receptors for epidermal growth factor, insulin-like growth factor-1, and platelet-derived growth factor can be transactivated through GPCRs to give proliferative and MAP kinase responses. Also, signalling through dopamine receptors leads to platelet-derived growth factor activation that results in changes in synaptic transmission [267].

Activation of TrkA and TrkB receptors can also occur *via* a GPCR mechanism, in the absence of NGF or BDNF [268,269]. Two GPCR ligands, adenosine and pituitary adenylate cyclase-activating peptide (PACAP) can activate Trk receptor activity to increase the survival of neural cells through stimulation of Akt activity. The effects of adenosine and PACAP can be blocked by K252a, an inhibitor of Trk tyrosine kinases. In contrast to neurotrophin treatment, Trk receptor activation by adenosine analogues and PACAP is sensitive to transcriptional and translational inhibitors, with the majority of the transactivated Trk receptors found in intracellular membranes, suggesting that receptor signalling may occur and persist inside of neuronal cells [270]. More recent studies indicate that Fyn is activated by GPCR stimulation and is responsible for transactivation of Trk receptors on intracellular membranes [271].



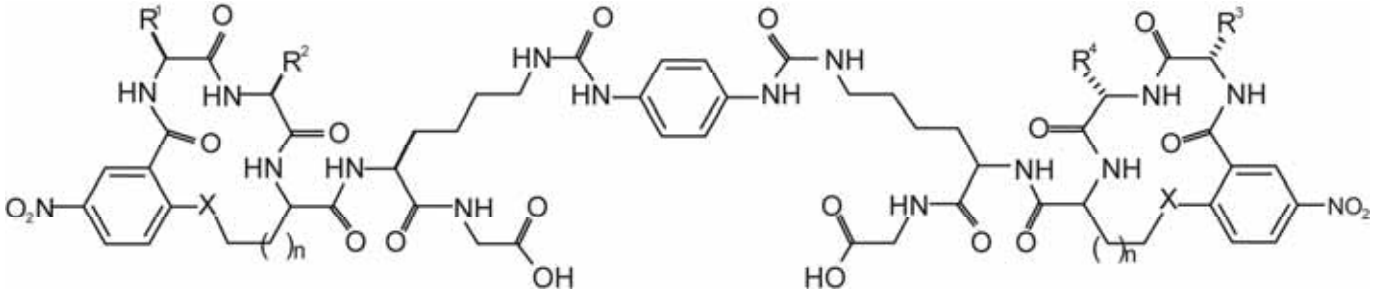


Fig. (8). TrkC agonist structure. [Reprinted from *Bioorganic & Medicinal Chemistry Letters, 11(2),* A.J. Zhang, S. Khare, K. Gokulan, D.S. Linthicum and K. Burgess, Dimeric β-turn peptidomimetics as ligands for the neurotrophin receptor TrkC, 207-210 (Scheme 1, structure 1), Copyright (2001), with permission from Elsevier].

Fig. **(9)**. The dimeric HAVDI peptide SW4 attenuates glutamate-induced toxicity in culture hippocampal neurons. Cultures (days *in vitro* 7–9) were pretreated with the indicated concentrations of SW4 peptide for 48 h and then exposed to 50 μM glutamate, and neuronal survival was quantified 24 h later by a colourimetric reaction (MTT). Data are means ± SD (three experiments) expressed relative to no glutamate (100%). $^{*}P < 0.05$ or $^{\#}P < 0.01$ *vs* glutamate only. [Reprinted from *Molecular and Cellular Neuroscience 26(1),* S.D. Skaper, L. Facci, G. Gilliams, E.-J. Williams, F.S. Walsh and P. Doherty, A dimeric version of the short N-cadherin binding motif HAVDI promotes neuronal cell survival by activating an N-cadherin/fibroblast growth factor receptor signalling cascade, 17-23 (Fig. **3**), Copyright (2004), with permission from Elsevier].

CONCLUDING REMARKS

Initially discovered as target-derived survival factors, the neurotrophins are known to control cell fate, axon growth and guidance, dendrite structure and pruning, synapse formation, and synaptic plasticity. Considerable progress has been made in defining the nature of neurotrophin receptors and in characterising their signalling pathways. Determining the pathways activated by distinct factors provides insight into the molecular basis by which specific and biological effects can be elicited. The Ras-MAP kinase-ERK pathway is critical in neurotrophin-induced differentiation responses, particularly in neuritogenesis. The multicomponent pathway leads from the Trk receptors to nuclear transcription factors. Ras-independent pathways are less well understood, but appear to be critical in the effects of neurotrophins on survival. The PI 3-kinase signalling pathway has been implicated in intracellular targeting and the survival response, while activation of PLC-γ1 results in activation of Ca^{2+} -regulated pathways that promote synaptic plasticity. Although exogenously given neurotrophic proteins are efficacious in animal models of nervous system injury, their general clinical usefulness remains in doubt. Small molecule mimetics may provide a viable alternative strategy A detailed understanding of neurotrophic factor-receptor interactions and intracellular signalling is crucial to the design of peptidyl and non-peptidyl small molecule neurotrophin mimetics which interact directly with the receptor, as illustrated for functional mimetics of neurotrophins acting at Trk receptors. Transactivation of Trks by GPCRs has emerged as a new theme in the biology of neurotrophin function. Although the precise role of transactivation is unknown, if small molecule GPCR ligands can ameliorate neuronal cell loss through Trk, transactivation may offer a new strategy for promoting trophic effects during neurodegeneration.

ABBREVIATIONS

BDNF	=	Brain-derived neurotrophic factor
CNS	=	Central nervous system
DAB_{389}-NT4	=	A diphtheria toxin-neurotrophin-4/5 fusion protein
FGF	=	Fibroblast growth factor
GDNF	=	Glial cell line-derived neurotrophic factor
GPCR	=	G-protein-coupled receptor
MAP	=	Mitogen-activated protein
MEK	=	MAP-ERK kinase kinase
SOS	=	Son of sevenless
NCAM	=	Neural cell adhesion molecule
NGF	=	Nerve growth factor
NT-3	=	Neurotrophin-3
NT-4/5	=	Neurotrophin-4/5
PACAP	=	pituitary adenylate cyclase-activating peptide
PI 3-kinase	=	Phosphatidylinositol 3-kinase
PLC-γ1	=	phospholipase C-γ1
PNS	=	Peripheral nervous system
$p75^{NTR}$	=	Low affinity neurotrophin receptor
SH2	=	Src homology 2
Trk	=	Tropomyosin-related kinase

REFERENCES

[1] Levi-Montalcini, R.; Skaper, S.D.; Aloe, L.; Leon, A. In *Encyclopedia of Neuroscience,* Second Edition, Adelman, G.; Smith, B.H., Eds.; Elsevier Science B.V: Amsterdam, **1999**, pp. 1297-1299.
[2] Levi-Montalcini, R.; Hamburger, V. *J. Exp. Zool.,* **1951**, *116,* 321.
[3] Levi-Montalcini, R.; Hamburger, V. *J. Exp. Zool.,* **1953**, *123,* 233.
[4] Levi-Montalcini, R.; *Science,* **1987**, *237,* 1154.
[5] Cohen, S. *Proc. Natl. Acad. Sci. USA,* **1960**, *46,* 302.
[6] Levi-Montalcini, R.; Booker, B. *Proc. Natl. Acad. Sci. USA,* **1960**, *46,* 384.
[7] Thoenen, H. *Science,* **1995**, *270,* 593.
[8] Thoenen, H.; Barde, Y.-A. *Physiol. Rev.,* **1980**, *60,* 1284.
[9] Barde, Y.-A. *Neuron,* **1989**, *2,* 1525.
[10] Hamburger, V.; Brunso-Bechthold, J.K.; Yip, J.W. *J. Neurosci.,* **1981**, *1,* 60.
[11] Bothwell, M. *Annu. Rev. Neurosci.,* **1995**, *18,* 223.
[12] Otten, U.; Scully, J.L.; Ehrhard, P.B.; Gadient, R.A. *Prog. Brain Res.,* **1994**, *103,* 293.
[13] Horigome, K.; Pryor, E.D.; Bullock, E.M.; Johnson, E.M. Jr. *J. Biol. Chem.,* **1993**, *268,* 14881.
[14] Solomon, A.; Aloe, L.; Pe'er, J.; Frucht-Pery, J.; Bonini, S.; Levi-Schaffer, F. *J. Allergy Clin. Immunol.,* **1998**, *102,* 454.
[15] Pincelli, C.; Sevignani, C.; Manfredini, R.; Grande, A.; Fantini, F.; Bracci-Laudiero, L; Aloe, L.; Ferrari, S.; Cossarizza, A.; Giannnetti, A. *J. Invest. Dermatol.,* **1994**, *103,* 13.
[16] Yaar, M.; Eller, M.S.; DiBenedetto, P.; Reenstra, W.R.; Zhai, S.; McQuaid, T.; Archambault, M.; Gilchrest, B.A. *J. Clin. Invest.,* **1994**, *94,* 1550.
[17] Polak, M.; Scharfmann, R.; Seilheimer, B.; Eisenbarth, G.; Dressler, D.; Verma, I.M.; Potter, H. *Proc. Natl. Acad. Sci. USA,* **1993**, *90,* 5781.
[18] Meakin, S.O.; Shooter, E.M. *Trends Neurosci.,* **1992**, *15,* 323.
[19] Patterson, J.C.; Childs, G.V. *Endocrinology,* **1994**, *135,* 1697.
[20] Levi-Montalcini, R.; Skaper, S.D.; Dal Toso, R.; Petrelli, L.; Leon, A. *Trends Neurosci.,* **1996**, *19,* 514.
[21] Skaper, S.D. *Mol. Neurobiol.,* **2001**, *24,* 183.
[22] Sofroniew, M.V.; Howe, C.L.; Mobley, W.C. *Annu. Rev. Neurosci.,* **2001**, *24,* 1217.
[23] Snider, W.D. *Cell,* **1994**, *77,* 627.
[24] Yuen, E.C.; Mobley, W.C. *Ann. Neurol.,* **1996**, *40,* 346.
[25] Jonsson, E.; Brene, S., Zhang, X.R.; Nimgaonkar, V.L.; Tylec, A.; Schalling, M.; Sedvall, G. *Acta Psychiatr. Scand.,* **1997**, *95,* 414.
[26] Thome, J.; Foley, P.; Riederer, P. *J. Neural Transm.,* **1998**, *105,* 85.
[27] Oppenheim, R.W. *Annu. Rev. Neurosci*., **1991**, 14, 451.
[28] Purves, D. *Body and Brain. A Trophic Theory of Neural Connections,* Harvard University Press: Cambridge, MA. **1988**.
[29] Barde, Y.-A.; Edgar, D.; Thoenen, H. *EMBO. J.,* **1982**, *1,* 549.
[30] Leibrock, J.; Lottspeich, F.; Hohn, A.; Hofer, M.; Hengerer, B.; Masiakowski, P.; Theonen, H.; Barde, Y.-A. *Nature,* **1994**, *341,* 149.

[31] Ernfors, P.; Ibáñez, C.F.; Ebendal, T.; Olson, L.; Persson, H. *Proc. Natl. Acad. Sci. USA,* **1990**, *87,* 5454.

[32] Hohn, A.; Leibrock, J.; Bailey, K.; Barde, Y.-A. *Nature,* **1990**, *344,* 339.

[33] Jones, K.R.; Reichardt, L.F. *Proc. Natl. Acad. Sci. USA,* **1990**, *87,* 8060.

[34] Kaisho, Y.; Yoshimura, K.; Nakahama, K. *FEBS Lett.,* **1990**, *266,* 187.

[35] Maisonpierre, P.C.; Belluscio, L.; Squinto, S.; Ip, N.Y.; Furth, M.E.; Lindsay, R.M.; Yancopoulos, G.D. *Science,* **1990**, *247,* 1446.

[36] Rosenthal, A.; Goeddel, D.V.; Nguyen, T.; Lewis, M.; Shih, A.; Laramee, G.R.; Nikolics, K.; Winslow, J.W. *Neuron,* **1990**, *4,* 767.

[37] Negro, A.; Tavella, A.; Grandi, C.; Skaper, S.D. *J. Neurochem.,* **1994**, *62,* 471.

[38] Berkemeier, L.R.; Winslow, J.W.; Kaplan, D.R.; Nicolics, K.; Goeddel, D.V.; Rosenthal, A. *Neuron,* **1991**, *7,* 857.

[39] Hallböök, F.; Ibáñez, C.F.; Persson, H. *Neuron,* **1991**, *6,* 845.

[40] Ip, N.Y.; Ibáñez, C.F.; Nye, S.H.; McClain, J.; Jones, P.F.; Gies, D.R.; Belluscio, L.; LeBeau, M.M.; Espinosa, R. III; Squinto, S.P.; Persson, H.; Yancopoulos, G.D. *Proc. Natl. Acad. Sci. USA,* **1992**, *89,* 3060.

[41] Ip, N.Y.; Stitt, T.N.; Tapley, P.; Klein, R.; Glass, D.J; Fandl, J.; Greene, L.A.; Barbacid, M.; Yancopoulos, G.D. *Neuron,* **1993**, *10,* 137.

[42] Götz, R.; Köster, R.; Winkler, C.; Raulf, F.; Lottspeich, F.; Schartl, M.; Thoenen, H. *Nature,* **1994**, *372,* 266.

[43] Lai, K.-O.; Fu, W.-Y.; Ip, F.C.F.; Ip, N.Y. *Mol. Cell. Neurosci.,* **1998**, *11,* 64.

[44] Eide, F.F.; Lowenstein, D.H.; Reichardt, L.F. *Exp. Neurol.,* **1993**, *121,* 200.

[45] Ibáñez, C.F. *Trends Biotech.,* **1995**, *13,* 227.

[46] DiCicco-Bloom, E.; Friedman, W.J.; Black, I.B. *Neuron,* **1993**, *11,* 1101.

[47] Verdi, J.M.; Anderson, D.J. *Neuron,* **1994**, *13,* 1359.

[48] Schober, A.; Wolf, N.; Huber, K.; Hertel, R.; Krieglstein, K.; Minichiello, L.; Kahane, N.; Widenfalk, J.; Kalcheim, C.; Olson, L.; Klein, R.; Lewin, G.R.; Unsicker, K. *J. Neurosci.,* **1998**, *18,* 7272.

[49] Roosen, A.; Schober, A.; Strelau, J.; Böttner, M.; Faulhaber, J.; Bendner, G.; McIlwrath, S.L.; Seller, H.; Ehmke, H.; Lewin, G.R.; Unsicker, K. *J. Neurosci.,* **2001**, *21,* 3073.

[50] Mannion, R.J.; Costigan, M.; Decosterd, I.; Amaya, F.; Ma, Q.-P.; Holstege, J.C.; Ji, R.-R.; Acheson, A.; Lindsay, R.M.; Wilkinson, G.A.; Woolf, C.J. *Proc. Natl. Acad. Sci. USA,* **1999**, *96,* 9385.

[51] Ip, N.Y.; Yancopoulos, G.D. *Annu. Rev. Neurosci.,* **1996**, *19,* 491.

[52] Lindsay, R.M.; Yancopoulos, G.D. *Neuron,* **1996**, *17,* 571.

[53] Doré, S., Kar, S.; Quirion, R. *Trends Neurosci.,* **1997**, *20,* 326.

[54] Walicke, P.; Cowan, W.M.; Ueno, N.; Baird, A.; Guillemin, R. *Proc. Natl. Acad. Sci. USA,* **1986**, *83,* 3012.

[55] Farkas, L.M.; Dunker, N.; Roussa, E.; Unsicker, K.; Krieglstein, K. *J. Neurosci.,* **2003**, *23,* 5178.

[56] Miao, N.; Wang, M.; Ott, J.A.; D-Alessandro, J.S.; Woolf, T.M.; Bumcrot, D.A.; Mahanthappa, K.; Pang, K. *J. Neurosci.,* **1997**, *17,* 5891.

[57] Hallböök, F. *Curr. Opin. Neurobiol.,* **1999**, *9,* 616.

[58] Lee, R.; Kermani, P.; Teng, K.K.; Hempstead, B.L. *Science,* **2001**, *294,* 1945.

[59] Skaper, S.D.; Pollock, M.; Facci, L. *Mol. Brain Res.,* **2001**, *97,* 177.

[60] Cotrina, M.L.; Gonzalez-Hoyuela, M.; Barbas, J.A.; Rodriguez-Tebar, A. *Dev. Biol.,* **2000**, *228,* 326.

[61] Friedman, W.J. *J. Neurosci.,* **2000**, *20,* 6340.

[62] Nykjaer, A.; Lee, R.; Teng, K.K.; Jansen, P.; Madsen, P.; Nielsen, M.S.; Jacobsen, C.; Kliemannel, M.; Schwarz, E.; Willnow, T.E.; Hempstead, B.L.; Petersen, C.M. *Nature,* **2004**, *427,* 843.

[63] Beattie, M.S.; Harrington, A.W.; Lee, R.; Kim, J.Y.; Boyce, S.L.; Longo, F.M.; Bresnahan, J.C.; Hempstead, B.L.; Yoon, S.O. *Neuron,* **2002**, *36,* 375.

[64] Harrington, A.W.; Leiner, B.; Blechschmitt, C.; Arevalo, J.C.; Lee, R.; Mörl, K.; Meyer, M.; Hempstead, B.L.; Yoon, S.O.; Giehl, K.M. *Proc. Natl. Acad. Sci. USA, 101,* **2004**, 6226.

[65] Fahnestock, M.; Michalski, B.; Xu, B.; Coughlin, M.D. *Mol. Cell. Neurosci.,* **2001**, *18,* 210.

[66] Volosin, M.; Song, W.; Almeida, R.D.; Kaplan, D.R.; Hempstead, B.L.; Friedman, W.J. *J. Neurosci.,* **2006**, *26,* 7756.

[67] Capsoni, S.; Cattaneo, A. *Cell. Mol. Neurobiol.,* **2006**, *26,* 619.

[68] McDonald, N.Q.; Lapatto, R.; Murray-Rust, J.; Gunning, J.; Wlodawer, A.; Blundell, T.L. *Nature,* **1991**, *354,* 411.

[69] Holland, D.R.; Cousens, L.S.; Meng, W.; Matthews, B.W. *J. Mol. Biol.,* **1994**, *239,* 385.

[70] Robinson, R.C.; Radziejewski, C.; Stuart, D.I.; Jones, E.Y. *Biochemistry,* **1995**, *34,* 4139.

[71] Butte, M.J.; Hwang, P.K.; Mobley, W.C.; Fletterick, R.J. *Biochemistry,* **1998**, *37,* 16846.

[72] Robinson, R.C.; Radziejewski, C.; Spraggon, G.; Greenwald, J.; Kostura, M.R.; Burtnick, L.D.; Stuart, D.I.; Choe, S.; Jones, E.Y. *Protein Sci.,* **1999**, *8,* 2589.

[73] Wiesmann, C.; Ultsch, M.H.; Bass, S.H.; Vos, A.M. de. *Nature,* **1999**, *324,* 461.

[74] Barde, Y.-A. *J. Neurobiol.,* **1994**, *25,* 1329.

[75] Kullander, K.; Carlson, B.; Hallböök, F. *J. Mol. Evol.,* **1997**, *45,* 311.

[76] Butte, M.J. *Cell. Mol. Life Sci.,* **2001**, *58,* 1003.

[77] Radeke, M.J.; Misko, T.P.; Hsu, C.; Herzenberg, L.A.; Shooter, E. *Nature,* **1987**, *325,* 593.

[78] Chao, M.V.; Hempstead, B.L. *Trends Neurosci.,* **1995**, *18,* 321.

[79] Rodriguez-Tébar, A.; Dechant, G.; Barde, Y.-A. *Neuron,* **1990**, *4,* 487.

[80] Rodriguez-Tébar, A.; Dechant, G.; Götz, R.; Barde, Y.-A. *EMBO. J.,* **1992**, *11,* 917.

[81] Liepinsh, E.; Ilag, L.L.; Otting, G.; Ibáñez, C.F. *EMBO. J.,* **1997**, *16,* 4999.

[82] He, X.E.; Garcia, K.C. *Science,* **2004**, *304,* 870.

[83] Murray, S.S.; Perez, P.; Lee, R.; Hempstead, B.L.; Chao, M.V. *J. Neurosci.,* **2004**, *24,* 2742.

[84] Kaplan, D.R.; Hempstead, B.L.; Martin-Zanca, D.; Chao, M.V.; Parada, L.F. *Science,* **1991**, *252,* 554.

[85] Kaplan, D.R.; Martin-Zanca, D.; Parada, L.F. *Nature,* **1991**, *350,* 158.

[86] Klein, R.; Jing, S.; Nanduri, V.; O'Rourke, E.; Barbacid, M. *Cell,* **1991**, *65,* 189.

[87] Klein, R.; Nanduri, V.; Jing, S.; Lambelle, F.; Tapley, P.; Bryant, S.; Cordon-Cardo, C.; Jones, K.R.; Reichardt, L.F.; Barbacid, M. *Cell,* **1991**, *66,* 395.

[88] Soppet, D.; Escandón, E.; Maragos, J.; Middlemas, D.S.; Reid, S.W.; Blair, J.; Burton, L.E.; Stanton, B.R.; *Kaplan*, D.R.; Hunter, T.; Nikolics, K.; Parada, L.F. *Cell,* **1991**, *65,* 895.

[89] Squinto, S.P.; Stitt, T.N.; Aldrich, T.H.; Davis, S.; Bianco, S.M.; Radziejewski, C.; Glass, D.J.; Masiakowski, P.; Furth, M.E.; Valenzuela, D.M.; DiStefano, P.S.; Yancopoulos, G.D. *Cell,* **1991**, *65,* 885.

[90] Lamballe, F.; Klein, R.; Barbacid, M. *Cell,* **1991**, *66,* 967.

[91] Cordon-Cardo, C.; Tapley, P.; Jing, S.; Nanduri, V.; O'Rourke, E.; Lamballe, F.; Kovary, K.; Klein, R.; Jones, K.R.; Reichardt, L.F.; Barbacid, M. *Cell,* **1991**, *66,* 173.

[92] Ultsch, M.H.; Wiesmann, C.; Simmons, L.C.; Henrich, J.; Yang, M.; Reilly, D.; Bass, S.H.; Vos, A.M. *J. Mol. Biol.,* **1999**, *290,* 149.

[93] Clary, D.O.; Reichardt, L.F. *Proc. Natl. Acad. Sci. USA,* **1994**, *91,* 11133.

[94] Strohmaier, C.; Carter, B.D.; Urfer, R.; Barde, Y.-A.; Dechant, G. *EMBO J.,* **1996**, *15,* 3332.

[95] Garner, A.S.; Menegay, H.J.; Boeshore, K.L.; Xie, X.Y.; Voci, J.M.; Johnson, J.E.; Large, T.H. *J. Neurosci.,* **1996**, *16,* 1740.

[96] Eide, F.F.; Vining, E.R.; Eide, B.L.; Zang, K.; Wang, X.Y.; Reichardt, L.F. *J. Neurosci.,* **1996**, *16,* 3123.

[97] Rose, C.R.; Blum, R.; Pichler, B.; Lepier, A.; Kafitz, K.W.; Konnerth, A. *Nature,* **2003**, *426,* 74.

[98] Guiton, M.; Gunn-Moore, F.J.; Glass, D.J.; Geis, D.R.; Yancopoulos, G.D.; Tavare, J.M. *J. Biol. Chem.,* **1995**, *270,* 20384.

[99] Du, J.; Feng, L.; Zaitsev, E.; Je, H.S.; Liu, X.W.; Lu, B. *J. Cell Biol.,* **2003**, *163,* 385.

[100] Benedetti, M.; Levi, A.; Chao, M.V. *Proc. Natl. Acad. Sci. USA,* **1993**, *90,* 7859.

[101] Bibel, M.; Hoppe, E.; Barde, Y.-A. *EMBO. J.,* **1999**, *18,* 616.

[102] Davies, A.M.; Lee, K.F.; Jaenisch, R. *Neuron,* **1993**, *11,* 565.

[103] Mahadeo, D.; Kaplan, L.; Chao, M.V.; Hempstead, B.L. *J. Biol. Chem.,* **1994**, *269,* 6884.

[104] Esposito, D.; Patel, P.; Stephens, R.M.; Perez, P.; Chao, M.V.; Kaplan, D.R.; Hempstead, B.L. *J. Biol. Chem.,* **2001**, *276,* 32687.

[105] Curtis, R.; Adryan, K.M.; Stark, J.L.; Park. J.S.; Compton, D.L.; Weskamp, G.; Huber, L.J.; Chao, M.V.; Jaenisch, R.; Lee, K.-F.; Lindsay, R.M.; DiStefano, P.S. *Neuron,* **1995**, *14,* 1201.

[106] Makkerh, J.P.; Ceni, C.; Auld, D.S.; Vaillancourt, F.; Dorval, G.; Barker, P.A. *EMBO. J.,* **2005**, *6,* 936.

[107] Geetha, T.; Jiang, J.; Wooten, M.W. *Mol. Cell,* **2005**, *20,* 301.
[108] Bentley, C.A.; Lee, K.F. *J. Neurosci.,* **2000**, *20,* 7706.
[109] Harrison, S.M.W.; Jones, M.E.; Uecker, S.; Albers, K.M.; Kudrycki, K.E.; Davis, B.M. *J. Comp. Neurol.,* **2000**, *424,* 99.
[110] Lee, K.-F.; Li, E.; Huber, L.J.; Landis, S.C.; Sharpe, A.H.; Chao, M.V.; Jaenisch, R. *Cell,* **1992**, *69,* 737.
[111] Lee, K.F.; Bachman, K.; Landis, S.; Jaenisch, R. *Science,* **1994**, *263,* 1447.
[112] Stucky, C.L.; Koltzenburg, M. *J. Neurosci.,* **1997**, *17,* 4398.
[113] Smeyne, R.J.; Klein, R.; Schnapp, A.; Long, L.K.; Bryant, S.; Lewin, A.; Lira, S.A.; Barbacid, M. *Nature,* **1994**, *368,* 246.
[114] Klein, R.; Smeyne, R.J.; Wurst, W.; Long, L.K.; Auerbach, B.A.; Joyner, A.L.; Barbacid, M. *Cell,* **1993**, *75,* 113.
[115] Conover, J.C.; Erickson, J.T.; Katz, D.M.; Bianchi, L.M.; Poueymirou, W.T.; McClain, J.; Pan, L.; Helgren, M.; Ip, N.Y.; Boland, P.; Friedman, B.; Wiegand, S.; Vejsada, R.; Kato, A.C.; DeChiara, T.M.; Yancopoulos, G.D. *Nature, 1995, 375,* 235.
[116] Liu, X.; Ernfors, P.; Wu, H.; Jaenisch, R. *Nature,* **1995**, *375,* 238.
[117] Wiley, R.G. *Trends Neurosci.,* **1992**, *15,* 285.
[118] Heckers, S.; Ohtake, T.; Wiley, R.G.; Lappi, D.A.; Geula, C.; Mesulam, M.-M. *J. Neurosci.,* **1994**, *14,* 1271.
[119] Negro, A.; Skaper, S.D. *J. Neurochem.,* **1997**, *68,* 554.
[120] Rudge, J.S.; Li, E.; Pasnikowski, E.M.; Mattsson, K.; Yancopoulos, G.D.; Wiegand, S.J.; Lindsay, R.M.; Ip, N.Y. *Eur. J. Neurosci.,* **1994**, *6,* 693.
[121] Lindholm, D.; Dechant, G.; Heisenberg, C.-P.; Thoenen, H. *Eur. J. Neurosci.,* **1993**, *5,* 1455.
[122] Gao, W.-Q.; Zheng, J.L.; Karihaloo, M. *J. Neurosci.,* **1995**, *15,* 2656.
[123] Kahle, P.; Burton, L.E.; Schmelzer, C.H.; Hertel, C. *J. Biol. Chem.,* **1992**, *267,* 22707.
[124] Ibáñez, C.F.; Ilag, L.L.; Murray-Rust, J.; Perrson, H. *EMBO J.,* **1993**, *12,* 2281.
[125] Shih, A.; Laramee, G.R.; Schmelzer, C.H.; Burton, L.E.; Winslow, J.W. *J. Biol. Chem.,* **1994**, *269,* 27679.
[126] Weiss, A.; Schlessinger, J. *Cell,* **1998**, *94,* 277.
[127] Huang, E.J.; Reichardt, L.F. *Annu. Rev. Biochem.,* **2003**, *72,* 609.
[128] Segal, R.A.; Greenberg, M.E. *Annu. Rev. Neurosci.,* **1996**, *19,* 463.
[129] York, R.D.; Molliver, D.C.; Grewal, S.S.; Stenberg, P.E.; McCleskey, E.W.; Stork, P.J.S. *Mol. Cell. Biol.,* **2000**, *20,* 8069.
[130] Ng, N.F.L.; Shooter, E.M. *J. Biol. Chem.,* **1993**, *268,* 25329.
[131] Borasio, G.D.; Markus, A.; Wittinghofer, A.; Barde, Y.-A.; Heumann, R. *J. Cell Biol.,* **1993**, *121,* 665.
[132] Nobes, C.D.; Tolkovsky, A.M. *Eur. J. Neurosci.,* **1995**, *7,* 344.
[133] Vogel, K.S.; Brannan, C.I.; Jenkins, N.A.; Copeland, N.G.; Parada, L.F. *Cell,* **1995**, *82,* 733.
[134] Nimnual, A.S.; Yatsula, B.A.; Bar-Sagi, D. *Science,* **1998**, *279,* 560.
[135] Robinson, K.N.; Manto, K.; Buchsbaum, R.J.; MacDonald, J.I.; Meakin, S.O. *J. Biol. Chem.,* **2005**, *280,* 225.
[136] Wood, K.W.; Sarnecki, C.; Roberts, T.M.; Blenis, J. *Cell,* **1992**, *68,* 1041.
[137] Oshima, M.; Sithanandam, G.; Rapp, U.; Guroff, G. *J. Biol. Chem.,* **1993**, *266,* 23753.
[138] Lange-Carter, C.A.; Johnson, G.L. *Science,* **1994**, *265,* 1458.
[139] Xing, J.; Kornhauser, J.M.; Xia, Z.; Thiele, E.A.; Greenberg, M.E. *Mol. Cell. Biol.,* **1998**, *18,* 1955.
[140] Crews, C.; Alessandrini, A.; Erikson, E. *Science,* **1992**, *258,* 478.
[141] Bar-Sagi, D.; Feramisco, J.R. *Cell,* **1985**, *42,* 841.
[142] Wood, K.W.; Qi, H.; D'Arcangelo, G.; Armstrong, R.C.; Roberts, T.M.; Halegoua, S. *Proc. Natl. Acad. Sci. USA,* **1993**, *90,* 5016.
[143] Cowley, S.; Paterson, H.; Kemp, P.; Marshall, C. *Cell,* **1994**, *77,* 841.
[144] Szeberenyi, J.; Cai, H.; Cooper, G. *Mol. Cell. Biol.,* **1990**, *10,* 5324.
[145] Schaap, D.; van der Wal, J.; Howe, L.R.; Marshall, C.J.; van Blitterswijk, W.J. *J. Biol. Chem.,* **1993**, *268,* 20232.
[146] Alessi, D.R.; Cuenda, A.; Cohen, P.; Dudley, D.T.; Saltiel, A.R. *J. Biol. Chem.,* **1995**, *270,* 27489.
[147] Skaper, S.D.; Walsh, F.S. *Mol. Cell. Neurosci.,* **1998**, *12,* 179.
[148] Creedon, D.J.; Johnson, E.M. Jr.; Lawrence, J.C. Jr. *J. Biol. Chem.,* **1996**, *271,* 20713.
[149] Ouwens, D.M.; de Ruiter, N.D.; van der Zon, G.C.M.; Carter, A.P.; Schouten, J.; van der Burgt, C.; Kooistra, K.; Bos, J.L.; Maassen, J.A.; van Dam, H. *EMBO J.,* **2002**, *21,* 3782.
[150] Shi, G.-X.; Andres, D.A. *Mol. Cell. Biol.,* **2005**, *25,* 830.
[151] Shi, G.X.; Han, J.; Andres, D.A. *J. Biol. Chem.,* **2005**, *37,* 37599.
[152] Watson, F.L.; Heerssen, H.M.; Bhattacharyya, A.; Klesse, L.; Lin, M.Z.; Segal, R.A. *Nat. Neurosci.,* **2001**, *4,* 981.
[153] Xu, B.-E.; Stippec, S.; Lenertz, L.; Lee, B.-H.; Zhang, W.; Lee, Y.-K.; Cobb, M.H. *J. Biol. Chem.,* **2004**, *279,* 7826.
[154] Horstmann, S.; Kahle, P.J.; Borasio, G.D. *J. Neurosci. Res.,* **1998**, *52,* 483.
[155] Pearson, G.; Robinson, F.; Gibson, T.B.; Xu, B.-E.; Karandikar, M.; Berman, K.; Cobb, M.H. *Endocr. Rev.,* **2001**, *22,* 153.
[156] Hill, C.; Treisman, R. *Cell,* **1995**, *80,* 199.
[157] Marinissen, M.J.; Chiariello, M.; Pallante, M.; Gutkind, J.S. *Mol. Cell. Biol.,* **1999**, *19,* 4289.
[158] Gille, H.; Sharrocks, A.D.; Shaw, P.E. *Nature,* **1992**, *358,* 414.
[159] Douville, E.; Downward, J. *Oncogene,* **1997**, *15,* 373.
[160] Virdee, K.; Tolkovsky, A.M. *Eur. J. Neurosci.,* **1995**, *7,* 2159.
[161] Virdee, K.; Tolkovsky, A.M. *J. Neurochem.,* **1996**, *67,* 1801.
[162] Kauffmann-Zeh, A.; Rodriguez-Viciana, P.; Ulrich, E.; Gilbert, C.; Coffer, P.; Downward, J.; Evan, G. *Nature, 1997, 385,* 544.
[163] Vaillant, A.R.; Mazzoni, I.; Tudan, C.; Boudreau, M.; Kaplan, D.R.; Miller, F.D. *J. Cell Biol.,* **1999**, *146,* 955.
[164] Tolkovsky, A. *Trends Neurosci.,* **1997**, *20,* 1.
[165] Holgado-Madruga, M.; Moscatello, D.K.; Emlet, D.R.; Dieterich, R.; Wong, A.J. *Proc. Natl. Acad. Sci. USA,* **1997**, *94,* 12419.
[166] Yamada, M.; Ohnishi, H.; Sano, S.; Nakatani, A.; Ikeuchi, T.; Hatanaka, H. *J. Biol. Chem.,* **1997**, *272,* 30334.
[167] Yao, R.; Cooper, G.M. *Science,* **1995**, *267,* 2003.
[168] D'Mello, S.R.; Borodezt, K.; Soltoff, S.P. *J. Neurosci.,* **1997**, *17,* 1548.
[169] Skaper, S.D.; Floreani, M.; Negro, A.; Facci, L.; Giusti, P. *J. Neurochem.,* **1998**, *70,* 1859.
[170] Marte, B.M.; Downward, J. *Trends Biochem. Sci.,* **1997**, *22,* 355.
[171] Chung, J.; Grammer, T.; Lemon, K.; Kazlauskas, A.; Blenis, J. *Nature,* **1994**, *370,* 1.
[172] Yuan, J.; Lipinski, M.; Degterev, A. *Neuron,* **2003**, *40,* 401.
[173] Brunet, A.; Datta, S.R.; Greenberg, M.E. *Curr. Opin. Neurobiol.,* **2001**, *11,* 297.
[174] Dudek, H.; Datta, S.R.; Franke, T.F.; Birnbaum, M.J.; Yao, R.; Cooper, G.M.; Segal, R.A.; Kaplan, D.R.; Greenberg, M.E. *Science,* **1997**, *275,* 661.
[175] Philpott, K.L.; McCarthy, M.J.; Klippel, A.; Rubin, L.L. *J. Cell Biol.,* **1997**, *139,* 809.
[176] Yuan, X.-B.; Jin, M.; Xu, X.; Song, Y.-Q.; Wu, C.-P.; Poo, M.-M.; Duan, S. *Nat. Cell Biol.,* **2003**, *5,* 38.
[177] Corbit, K.C.; Foster, D.A.; Rosner, M.R. *Mol. Cell. Biol.,* **1999**, *19,* 4209.
[178] Matsumoto, T.; Numakawa, T.; Adachi, N.; Yokomaku, D.; Yamagishi, S.; Takei, N.; Hatanaka, H. *J. Neurochem.,* **2001**, *79,* 522.
[179] Minichiello, L.; Calella, A.M.; Medina, D.L.; Bonhoeffer, T.; Klein, R.; Korte, M. *Neuron,* **2002**, *36,* 121.
[180] Rabin, S.; Cleghorn, V.; Kaplan, D. *Mol. Cell. Biol.,* **1993**, *13,* 2203.
[181] Peng, X.; Greene, L.; Kaplan, D.; Stephens, R. *Neuron,* **1995**, *15,* 395.
[182] Yano, H.; Ninan, I.; Zhang, H.; Milner, T.A.; Arancio, O.; Chao, M.V. *Nat. Neurosci.,* **2006**, *9,* 1009.
[183] Lu, B. *Learn. Mem.,* **2003**, *10,* 86.
[184] Huang, C.-S.; Zhou, J.; Feng, A.K.; Lynch, C.C.; Klumperman, J.; DeArmond, S.J.; Mobley, W.C. *J. Biol. Chem.,* **1999**, *274,* 36707.
[185] Shao, Y.; Akmentin, W.; Toledo-Aral, J.J.; Rosenbaum, J.; Valdez, G.; Cabot, J.B.; Hilbush, B.S.; Halegoua, S. *J. Cell Biol.,* **2002**, *157,* 679.
[186] Delcroix, J.D.; Valletta, J.S.; Wu, C.; Hunt, S.J.; Kowal, A.S.; Mobley, W.C. *Neuron,* **2003**, *39,* 69.
[187] Arévalo, J.C.; Waite, J.; Rajagopal, R.; Beyna, M.; Chen, Z.-Y.; Lee, F.S.; Chao, M.V. *Neuron,* **2006**, *50,* 549.
[188] Tsui-Pierchala, B.A.; Encinas, M.; Milbrandt, J.; Johnson, Jr. E.M. *Trends Neurosci.,* **2002**, *25,* 412.
[189] Suzuki, S.; Numakawa, T.; Shimazu, K.; Koshimizu, H.; Hara, T.; Hatanaka, H.; Mei, L.; Lu, B.; Kojima, M. *J. Cell Biol.,* **2004**, *167,* 1203.
[190] Pereira, D.B.; Chao, M.V. *J. Neurosci.,* **2007**, *27,* 4859.
[191] Hefi, F. *J. Neurosci.,* **1986**, *6,* 2155.
[192] Williams, L.R.; Varon, S.; Peterson, G.M.; Wictorin, K.; Fischer, W.; Björklund, A.; Gage, F.H. *Proc. Natl. Acad. Sci. USA,* **1986**, *83,* 9231.

[193] Li, Y.; Holtzman, D.M.; Kromer, L.F.; Kaplan, D.R.; Chua-Couzens, J.; Clary, D.O.; Knusel, B.; Mobley, W.C. *J. Neurosci.,* **1995**, *15,* 2888.

[194] Mufson, E.J.; Kroin, J.S.; Sendera, T.J.; Sobreviela, T. *Prog. Neurobiol.,* **1999**, *57,* 451.

[195] Koliatsos, V.E.; Nauta, H.J.; Clatterbuck, R.E.; Holtzman, D.M.; Mobley, W.C.; Price, D.L. *J. Neurosci.,* **1990**, *10,* 3801.

[196] Smith, D.E.; Roberts, J.; Gage, F.H.; Tuszynski, M.H. *Proc. Natl. Acad. Sci. USA,* **1999**, *96,* 10893.

[197] Fischer, W.; Wictorin, K.; Bjorklund, A.; Williams, L.R.; Varon, S.; Gage, F.H. *Nature,* **1987**, *329,* 65.

[198] Markowska, A.L.; Koliatsos, V.E.; Breckler, S.J.; Price, D.L.; Olton, D.S. *J. Neurosci.,* **1994**, *14,* 4815.

[199] Kordower, J.H.; Emborg, M.E.; Bloch, J.; Ma, S.Y.; Chu, Y.; Leventhal, L.; McBride, J.; Chen, E.Y.; Palfi, S.; Roitberg, B.Z.; Brown, W.D.; Holden, J.E.; Pyzalski, R.; Taylor, M.D.; Carvey P.; Ling, Z.; Trono, D.; Hantraye, P.; Deglon, N.; Aebischer, P. *Science,* **2000**, *290,* 767.

[200] Björklund, A.; Rosenblad, C.; Winkler, C.; Kirik, D. *Neurobiol. Dis.,* **1997**, *4,* 186.

[201] Gash, D.M.; Zhang, Z.; Ovadia, A.; Cass, W.A.; Yi, A.; Simmerman, L.; Russell, D.; Martin, D.; Lapchak, P.A.; Collins, F.; Hoffer, B.J.; Gerhardt, G.A. *Nature,* **1996**, *380,* 252.

[202] Lange, D.J.; Felice, K.J.; Festoff, B.W.; Gawel, M.J.; Gelinas, D.F.; Kratz, R.; Lai, E.C.; Murphy, M.F.; Natter, H.M.; Norris, F.H.; Rudnicki, S. North American ALS/IGF-1 Study Group. *Neurology,***1996**, *47,* 93.

[203] Borasio, G.D.; Robberecht, W.; Leigh, P.N.; Emile, J.; Guiloff, R.J.; Jerusalem, F.; Silani, V.; Vos, P.E.; Wokke, J.H.; Dobbins, T. European ALS/IGF-1 Study Group. *Neurology,* **1999**, *51,* 583.

[204] The BDNF study group (Phase III). *Neurology,***1999**, *52,* 1427.

[205] Ochs, G.; Penn, R.D.; York, M.; Tonn, J.; Haigh, J.; Malta, E.; Traub, M.; Sendtner, M.; Toyka, K.V. *Amyotroph. Lateral Scler. Other Motor Neuron Disord.,* **2000**, *1,* 201.

[206] Beck, M.; Flachenecker, P.; Magnus, T.; Giess, R.; Reiners, K.; Toyka, K.V.; Naumann, M. *Amyotroph. Lateral Scler. Other Motor Neuron Disord.,* **2005**, *6,* 100.

[207] ALS CNTF Treatment Study Group. *Neurology,***1996**, *46,* 1244.

[208] Turner, M.R.; Parton, M.J.; Leigh, P.N. *Semin. Neurol.,* **2001**, *21,* 167.

[209] Nutt, J.G.; Burchiel, K.D.; Comella, C.L.; Jankovic, J.; Land, A.E.; Laws, Jr. E.R.; Lozano, A.M.; Penn, R.D.; Simpson, Jr. R.K.; Stacy, M.; Wooten, G.F.; and the ICV GDNF Study Group. *Neurology,* **2003**, *60,* 69.

[210] Lang, A.E.; Gill, S.; Patel, N.K.; Lozano, A.; Nutt, J.G.; Penn, R.; Brooks, D.J.; Hotton, G.; Moro, E.; Heywood, P.; Brodsky, M.A.; Burchiel, K.; Kelly, P.; Dalvi, A.; Scott, B.; Stacy, M.; Turner, D.; Wooten, V.G.F.; Elias, W.J.; Laws, E.R.; Dhawan, V.; Stoessl, A.J.; Matcham, J.; Coffey, R.J.; Traub, M. *Ann. Neurol.* **2006**, *59,* 459.

[211] Podulso, J.F.; Curran, G.L.; Gill, J.S. *J. Neurochem.,* **1998**, *71,* 1651.

[212] Wu, D.; Pardridge, W.M. *Proc. Natl. Acad. Sci. USA,* **1999**, *96,* 254.

[213] Zhang, Y.; Pardridge, W.M. *Stroke,* **2001**, *32,* 1378.

[214] Kotzbauer, P.T.; Lampe, P.A.; Heuckeroth, R.O.; Golden, J.P.; Creedon, D.J.; Johnson, Jr. E.M.; Milbrandt, J. *Nature,* **1996**, *384,* 467.

[215] Choi-Lundberg, D.L.; Lin, Q.; Chang, Y.-N.; Chiang, Y.L.; Hay, C.M.; Mohajeri, H.; Davidson, B.L.; Bohn, M.C. *Science,* **1997**, *275,* 838.

[216] Fjord-Larsen, L.; Johansen, J.L.; Kusk, P.; TornØe, J.; GrØnborg, M.; Rosenblad, C.; Wahlberg, L.U. *Exp. Neurol.,* **2005**, *195,* 49.

[217] Zhou, L.; Baumgartner, B.J.; Hill-Felberg, S.J.; McGowen, L.R.; Shine, H.D. *J. Neurosci.,* **2003**, *23,* 1424.

[218] Blesch, A.; Yang, Weidner, N.; Hoang, A.; Otero, D. *Mol. Cell. Neurosci.,* **2004**, *27,* 190.

[219] Taylor, L.; Jones, L.; Tuszynski, M.H.; Blesch, A. *J. Neurosci.,* **2006**, *26,* 9713.

[220] Tuszynski, M.H.; Thal, L.; Pay, M.; Salmon, D.P.; U, H.S.; Bakay, R.; Patel, P.; Blesch, A.; Vahlsing, H.L.; Ho, G.; Tong, G.; Potkin, S.G.; Fallon, J.; Hansen, L.; Mufson, E.J.; Kordower, J.H.; Gall, C.; Conner, J. *Nat. Med.,* **2005**, *11,* 551.

[221] Brewster, W.J.; Fernyhough, P.; Diemel, LT.; Mohiuddin, L.; Tomlinson, R.R. *Trends Neurosci.,* **1994**, *17,* 321.

[222] Anand, P.; Terenghi, G.; Warner, G.; Kopelman, P.; Williams-Chestnut, R.E.; Sinicropi, D.V. *Nat. Med.,* **1996**, *2,* 703.

[223] Apfel, S.C. *Rev. Neurobiol.,* **2002**, *50,* 393.

[224] Apfel, S.C.; Kessler, J.A.; Adornato, B.T.; Litchy, W.J.; Sanders, C.; Rask, C.A. NGD Study Group. *Neurology,* **1998**, *51,* 695.

[225] McArthur, J.C.; Yiannoutsos, C.; Simpson, D.M.; Adornato, B.T.; Singer, E.J.; Hollander, H.; Marra, C.; Rubin, M.; Cohen, B.A.; Tucker, T.; Navia, B.A.; Schifitto, G.; Katzenstein, D.; Rask, C.; Zaborski, L.; Smith, M.E.; Shriver, S.; Millar, M.; Clifford, D.B. *Neurology,* **2000**, *54,* 1080.

[226] Apfel, S.C.; Schwartz, S.; Adornato, B.T.; Freeman, R.; Biton, V.; Rendell, M.; Vinik, A.; Giuliani, M.; Stevens, J.C.; Barbano, R.; Dyck, P.J. *J. Am. Med. Assoc.,* **2000**, *284,* 2215.

[227] Schmidt, R.E.; Dorsey, D.A.; Beaudet, L.N.; Parvin, C.A.; Escandon, E. *J. Neuropathol. Exp. Neurol.,* **2001**, *60,* 263.

[228] Lewin, G.R.; Mendell, L.M. *Trends Neurosci.,* **1993**, *16,* 353.

[229] Calcutt, N.A.; Allendoerfer, K.L.; Mizisin, A.P.; Middlemas, A.; Freshwater, J.D.; Burgers, M.; Ranciato, R.; Delcroix, J.-D.; Taylor, F.R.; Shapiro, R.; Strauch, K.; Dudek, H.; Engber, T.M.; Galdes, A.; Rubin, L.L.; Tomlinson, D.R. *J. Clin. Invest.,* **2003**, *111,* 507.

[230] Mizisin, A.P.; Vu, Y.; Shuff, M.; Calcutt, N.A. *Diabetes,* **2004**, *53,* 1807.

[231] Saragovi, H.U.; Greene, M.I.; Chrusciel, R.A.; Kahn, M. *Biotechnology,* **1992**, *10,* 773.

[232] Wells, J.A. *Science,* **1996**, *273,* 449.

[233] Saragovi, H.U.; Greene, M.I.; Chrusciel, R.A.; Kahn, M. *Biotechnology,* **1992**, *10,* 773.

[234] Williams, W.V.; Moss, D.A.; Kieber-Emmons, T.; Cohen, J.A.; Myers, J.N.; Weiner, D.B.; Greene, M.I. *Proc. Natl. Acad. Sci. USA,* **1989**, *86,* 5537.

[235] Williams, W.V.; Kieber-Emmons, T.; VonFeldt, J.; Greene, M.I.; Weiner, D.B. *J. Biol. Chem.,* **1991**, *266,* 5182.

[236] McDowell, R.S.; Gadek, T.R. *J. Am. Chem. Soc.,* **1992**, *114,* 9245.

[237] Dechantsreiter, M.A.; Planker, E.; Mathä, B.; Lohof, E.; Hölzemann, G.; Jonczyk, A.; Goodman, S.L.; Kessler, H. *J. Med. Chem.,* **1999**, *42,* 3033.

[238] Saragovi, H.U.; Fitzpatrick, D.; Raktabuhr, A.; Nakanishi, H.; Kahn, M.; Green, M.I. *Science,* **1991**, *253,* 792.

[239] LeSauteur, L.; Wei, L.; Gibbs, B.F.; Saragovi, H.U. *J. Biol. Chem.,* **1995**, *270,* 6564.

[240] LeSauteur, L.; Cheung, N.-K.V.; Lisbona, R.; Saragovi, H.U. *Nat. Biotech.,* **1996**, *14,* 1120.

[241] Debeir, T.; Saragovi, H.U.; Cuello, A.C. *Proc. Natl. Acad. Sci. USA,* **1999**, 96, 4067.

[242] LeSauteur, L.; Maliartchouk, S.; Le Jeune, H.; Quirion, R.; Saragovi, H.U. *J. Neurosci.,* **1996**, *16,* 1308.

[243] Maliartchouk, S.; Debeir, T.; Beglova, N.; Cuello, A.C.; Gehring, K.; Saragovi, H.U. *J. Biol. Chem.,* **2000**, *275,* 9946.

[244] Xie, Y.; Tisi, M.A.; Yeo, T.T.; Longo, F.M. *J. Biol. Chem., 275,* **2000**, 29868.

[245] Maliartchouk, S.; Feng, Y.; Ivanisevic, L.; Debeir, T.; Cuello, A.C.; Burgess, K.; Saragovi, H.U. *Mol. Pharmacol.,* **2000**, *57,* 385.

[246] Rashid, K.; Van der Zee, C.E.E.M.; Ross, G.M.; Chapman, C.A.; Stanisz, J.; Riopelle, R.J.; Racine, R.J.; Fahnestock, M. *Proc. Natl. Acad. Sci. USA,* **1995**, *92,* 9495.

[247] Bruno, M.A.; Clarke, P.B.S.; Seltzer, A.; Quirion, R.; Burgess, K.; Cuello, A.C.; Saragovi, H.U. *J. Neurosci.,* **2004**, *24,* 8009.

[248] O'Leary, P.D.; Hughes, R.A. *J. Biol. Chem.,* **2003**, *278,* 25738.

[249] O'Leary, P.D.; Hughes, R.A. *J. Neurochem.,* **1998**, *70,* 1712.

[250] Zhang, A.J.; Khare, S.; Gokulan, K.; Linthicum, D.S.; Burgess, K. *Bioorg. Med. Chem. Lett.,* **2003**, *11,* 207.

[251] Williams, G.; Williams, E.-J.; Maison, P.; Pangalos, M.N.; Walsh, F.S.; Doherty, P. *J. Biol. Chem.,* **2005**, *280,* 5862.

[252] Skaper S.D. *Ann. NY Acad. Sci.,* **2005**, *1053,* 376.

[253] Williams, E.J.; Furness, J.; Walsh, F.S.; Doherty, P. *Neuron, **1994**, 13,* 583.

[254] Williams, E.J.; Walsh, F.S.; Doherty, P. J. *Neurochem.,* **1994**, *62,* 1231.

[255] Saffell, J.L.; Williams, E.J.; Mason, I.J.; Walsh, F.S.; Doherty, P. *Neuron,* **1997**, *18,* 231.

[256] Walmod, P.S.; Kolkova, K.; Berezin, V.; Bock, E. *Neurochem. Res.*, **2004**, *29,* 2015.

[257] Rønn, L.C.B.; Olsen, M.; Østergaard, S.; Kiselyov, V.; Berezin, V.; Mortensen, M.T.; Lerche, M.H.; Jensen, P.H.; Soroka, V.; Saffells,

J.L.; Doherty, P.; Poulsen, F.M.; Bock, E.; Holm, A. *Nat. Biotech.,* **1999**, *17,* 1000.
[258] Berezin, V.; Bock, E. *J. Mol. Neurosci.,* **2004**, *22,* 33.
[259] Rønn, L.C.B.; Doherty, P.; Holm, A.; Berezin, V.; Bock, E. *J. Neurochem.,* **2000**, *75,* 665.
[260] Williams, E.; Williams, G.; Gour, B.J.; Blaschuk, O.W.; Doherty, P. *J. Biol. Chem.,* **2000**, *275,* 4007.
[261] Williams, G.; Williams, E.-J.; Doherty, P. *J. Biol. Chem.,* **2000**, *277,* 4361.
[262] Ditlevsen, D.K.; Køhler, L.B.; Pedersen, M.V.; Risell, M.; Kolkova, K.; Meyer, M.; Berezin, V.; Bock, E. *J. Neurochem.,* **2003**, **84**, 546,
[263] Ditlevsen, D.K.; Berezin, V.; Bock, E. *Eur. J. Neurosci.,* **2007**, *25,* 1678.
[264] Skaper, S.D.; Facci, L.; Williams, G.; Williams, E.-J.; Walsh, F.S.; Doherty, P. *Mol. Cell. Neurosci.,* **2004**, *26,* 17.
[265] Daub, H.; Weiss, F.U.; Wallasch, C.; Ullrich, A. *Nature,* **1996**, *379,* 557.
[266] Luttrell, L.; Daaka, Y.; Lefkowitz, R. *Curr. Opin. Cell Biol.,* **1999**, *11,* 177.
[267] Kotecha, S.A.; Oak, J.N.; Jackson, M.F.; Perez, Y.; Orser, B.A.; Van Tol, H.H.; MacDonald, J.F. *Neuron,* **2002**, *35,* 1111.
[268] Lee, F.S.; Chao, M.V. *Proc. Natl. Acad. Sci. USA,* **2001**, *92,* 3555.
[269] Lee, F.S.; Rajagopal, R.; Kim, A.H.; Chang, P.C.; Chao, M.V. *J. Biol. Chem.,* **2002**, *277,* 9096.
[270] Rajagopal, R.; Chen, Z.-Y.; Lee, F.S.; Chao, M.V. *J. Neurosci.,* **2004**, *24,* 6650.
[271] Rajagopal, R.; Chao, M.V. *Mol. Cell. Neurosci.,* **2006**, *33,* 36.

Received: January 10, 2008 **Accepted: January 22, 2008**